A Project Report on

Sentiment – Augmented Stock Price Forecasting

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CERTIFICATE

This is to certify that the project entitled "Sentiment – Augmented Stock Price Forecasting" is a bonafide work of **Manasi Patil (22106042), Samiksha Patil (22106041), Smith Patil (23206010)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning**).

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Project Report Approval for B.E.

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Abstract

Stock price prediction remains a challenging task due to the complex and dynamic nature of financial markets, which are influenced by both historical trends and external factors such as news sentiment and investor behavior. This project presents an integrated stock market decision-support framework that combines price forecasting, sentiment analysis, and personalized recommendation.

The proposed system employs a Long Short-Term Memory (LSTM) network for time-series prediction, with its hyperparameters optimized using a Genetic Algorithm (GA) to improve model accuracy and convergence. Historical stock data obtained from yfinance is used for training the predictive model. In parallel, sentiment analysis is performed using FinBERT, a domain-adapted transformer model fine-tuned on financial text, to classify real-time news data retrieved from the Finnhub API into positive, negative, and neutral sentiments.

In addition, a profile-driven recommendation system is developed to generate personalized investment suggestions based on user inputs such as age, investment amount, and market type. Experimental evaluation on Amazon (AMZN) stock data demonstrates that the GA-optimized LSTM model outperforms the baseline model in terms of prediction accuracy and stability. The integration of sentiment insights further enhances contextual understanding, while the recommendation module ensures interpretability and user-centric decision support.

Overall, the proposed framework highlights the effectiveness of combining deep learning, natural language processing, and rule-based systems to deliver practical and reliable investment guidance for retail investors.

Keywords: Stock Price Prediction, LSTM, Genetic Algorithm, FinBERT, Sentiment Analysis, Recommendation System, Time Series Forecasting, Financial Markets

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ABBREVIATIONS

OHLCV	*Open, High, Low, Close, Volume*
RNN	*Recurrent Neural Network*
LSTM	*Long Short Term Memory*
GA	*Genetic Algorithm*
NLP	*Natural Language Processing*
BERT	*Bidirectional Encoder Representations from Transformers*
FinBERT	*Financial Bidirectional Encoder Representations from Transformers*
RSI	*Relative Strength Index*
SMA	*Simple Moving Average*
EMA	*Exponential Moving Average*
MAPE	*Mean Absolute Percentage Error*
RMSE	*Root Mean Square Error*
MAE	*Mean Absolute Error*
MSE	*Mean Squared Error*

Chapter 1

Introduction

The global financial market is a highly dynamic and complex system characterized by volatility, non-linearity, and high-dimensional data interactions. Predicting stock price movements within such an environment has long been regarded as a significant challenge for investors, analysts, and researchers alike. Traditional statistical models, including Autoregressive Integrated Moving Average (ARIMA), often fail to capture the stochastic and chaotic nature of modern financial markets, which are influenced not only by historical price patterns but also by real-time information flow, macroeconomic factors, and investor sentiment.

In response to these limitations, this project proposes a Hybrid Intelligent Framework for Stock Price Forecasting and User-Centric Recommendation, integrating advanced techniques from Deep Learning, Evolutionary Computing, and Natural Language Processing (NLP). The objective is to move beyond conventional forecasting approaches and develop a comprehensive decision-support system capable of delivering both accurate predictions and personalized investment insights.

At the core of the proposed system lies a Long Short-Term Memory (LSTM) network, a specialized form of Recurrent Neural Network (RNN) designed to effectively model time-series data by capturing long-term dependencies and mitigating the vanishing gradient problem. Despite its advantages, the performance of LSTM models is highly sensitive to hyperparameter configuration

To address this challenge, the project incorporates a Genetic Algorithm (GA) for automated hyperparameter optimization. Inspired by the principles of natural selection and evolution, the GA systematically evolves candidate solutions by applying operations such as selection, crossover, and mutation. Each configuration is evaluated using performance metrics including Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE). This evolutionary optimization enables the system to identify an optimal LSTM architecture tailored to the specific characteristics of financial data, thereby significantly improving prediction accuracy.

While quantitative analysis forms the foundation of forecasting, market behavior is also strongly influenced by qualitative factors, particularly news and investor sentiment. To incorporate this dimension, the system integrates a financial sentiment analysis module powered by a fine-tuned FinBERT model. Unlike generic sentiment analysis tools, FinBERT is specifically designed to interpret financial text and understand domain-specific terminology.

A key contribution of this project is the development of a custom-labeled financial sentiment dataset comprising approximately 1,200 samples. Using the Loughran-McDonald Financial Word List as a baseline, a hybrid labeling approach was employed, combining dictionary-based methods with rigorous manual verification, to ensure high annotation accuracy. The dataset is balanced across Positive, Negative, and Neutral classes, enabling robust model training. By analyzing real-time financial news data, the sentiment engine generates actionable signals such as Buy, Hold, or Sell, thereby enhancing the predictive capability of the system.

The final component of the framework is a Rule-Based Recommendation System, designed to translate analytical outputs into practical investment advice. Recognizing that investment decisions are inherently user-specific, the system incorporates individual parameters such as age, investment capacity, risk tolerance, and market preferences. These inputs are combined with LSTM-GA forecasts, sentiment scores, and technical indicators like the Relative Strength Index (RSI) to generate personalized and context-aware recommendations.

In summary, this project presents a holistic and multi-layered approach to stock market analysis by combining predictive modeling, sentiment intelligence, and expert decision systems. By bridging the gap between raw financial data and actionable insights, the proposed framework aims to provide a robust, scalable, and user-centric solution for modern investment decision-making.

Chapter 2

Literature Survey

[1] Stock Price Prediction Using LSTM on Indian Share Market (Ghosh et al., 2019) This study investigates the application of Long Short-Term Memory (LSTM) networks to predict future company growth and share prices within the Indian stock market. By analyzing top companies across sectors like Banking, IT, and Pharma, the researchers established that firms within the same sector share similar inner dynamics and growth patterns. A key finding of the research is that the model's predictive error decreases drastically when trained on longer spans of historical data (e.g., three years) compared to shorter monthly spans. The authors conclude that LSTM is superior for this task because its gate mechanism effectively handles the non-linear, irregular patterns of time-series financial data.

[2] Hyperparameter Optimization of LSTM Network Models through Genetic Algorithm (Gorgolis et al., 2019) This paper addresses the challenge of fine-tuning Artificial Neural Networks by proposing a Genetic Algorithm (GA) approach for hyperparameter optimization. The authors focus specifically on LSTM networks, which are essential for sequence prediction in NLP but notoriously difficult to tune manually or through exhaustive grid searches. By applying a GA to variables such as hidden layer size, batch size, and number of timesteps, the researchers identified configurations that achieve higher accuracy than standard default settings. The study concludes that GA-tuned models systematically outperform the "medium-sized" configurations used by popular platforms like TensorFlow, offering a more efficient path to high model performance.

[3] FinBERT: A Pre-trained Financial Language Representation Model for Financial Text Mining (Liu et al., 2020) This study proposes a FinBERT model designed to address the lack of labeled training data in the financial field by utilizing unsupervised transfer learning and multi-task pre-training. The model was pre-trained on 61GB of financial corpora using six self-supervised tasks—such as sentence deshuffling and capitalization prediction—to capture unique domain knowledge and semantic information. Experimental results show that this version of FinBERT achieves state-of-the-art performance across diverse tasks, including financial Sentence Boundary Detection (SBD), Sentiment Analysis (SA), and Question Answering (QA). The authors emphasize that FinBERT's multi-task learning architecture provides stable and clear enhancements even when trained on smaller, simulated financial datasets.

[4] A systematic review of stock market prediction using machine learning and statistical techniques (Kumar et al., 2020) This systematic review examines 30 research papers to categorize the methodologies, performance parameters, and dominant datasets used in stock market forecasting. The authors identify Artificial Neural Networks (ANN) and Support Vector Machines (SVM) as the most precise and powerful predictive tools currently available in the financial market. The review highlights that while traditional statistical models like ARIMA are still used, they are increasingly being replaced or combined with deep learning architectures to handle noisy, non-linear data. The paper concludes that for future systems to be more accurate, they must integrate distinctive parameters like government policy decisions and consumer sentiment alongside historical trading data.

[5] Stock Prediction Based on GA Feature Selection and LSTM (Chen & Zhou, 2020) This paper proposes a two-stage stock prediction model that integrates a Genetic Algorithm (GA) for feature selection with an LSTM neural network. In the first stage, the GA is used to statistically rank the importance of 40 different stock factors, including market, technical, and financial indicators. In the second stage, the optimal combination of these factors is fed into an LSTM network, which is chosen for its superior ability to handle time-series data. Empirical studies using Chinese market datasets demonstrate that the GA-LSTM model outperforms baseline models such as Random Forest and DA-RNN, effectively reducing Mean Square Error (MSE) through optimized multi-factor inputs.

[6] LSTM BASED STOCK PRICE PREDICTION (Ahire et al., 2021) This project implements a stock price prediction system using a type of Recurrent Neural Network known

as Long Short-Term Memory (LSTM). The model is designed to overcome the limitations of traditional RNNs by using a gate mechanism—input, forget, and output gates—to associate memory with remote inputs over time. Using the Yahoo Finance API to collect historical data for major companies like Google and Apple, the authors developed a system that generates future prediction graphs based on past price movements. The paper emphasizes that while the stock market is inherently uncertain, LSTM's ability to retain intermediary context provides a highly beneficial tool for investors seeking reliable financial growth indicators.

[7] How can we use artificial intelligence for stock recommendation and risk management? (Gonzales & Hargreaves, 2022) This paper proposes a personalized decision support system that uses hierarchical clustering and collaborative filtering to provide tailored stock recommendations. By segmenting users into short, medium, and long-term trading styles, the system improves computational efficiency and ensures that recommendations align with individual risk appetites. The authors evaluate three models—k-Nearest Neighbor (kNN), SVD, and Association Rule Mining (ARM)—finding that kNN performs best across both predictive and classification metrics. Additionally, the system incorporates Value-at-Risk (VaR) through Monte Carlo simulations to support risk-informed decisions and minimize potential financial losses for investors.

[8] A sentiment analysis approach to the prediction of market volatility (Deveikyte et al., 2022) This research investigates the predictive power of sentiment extracted from news headlines, news stories, and Twitter on FTSE100 market volatility. The authors discovered a strong negative correlation (-0.7) between negative Twitter sentiment and next-day market volatility, noting that tweets are often timelier and more reactive than traditional news articles. By deploying topic modeling based on Latent Dirichlet Allocation (LDA) to extract feature vectors, the researchers developed a classifier capable of predicting directional volatility with 63% accuracy. The study concludes that while sentiment alone does not "cause" market changes in a Granger sense, the topics extracted from news and social media provide valuable signals for risk management and portfolio optimization.

[9] Machine learning approaches in stock market prediction: A systematic literature review (Mintarya et al., 2023) This systematic literature review (SLR) analyzes 30 studies to identify the prevailing trends and most effective machine learning models for stock market prediction. The research finds that Artificial Neural Networks (ANNs) are the most frequently used and accurate models due to their ability to comprehend complex, non-linear data trends.

The study tracks the evolution of ML in finance, noting that while interest peaked in 2013, the adoption of Long Short-Term Memory (LSTM) networks surged after 2015 as the use of Support Vector Machines (SVM) began to diminish. Ultimately, the authors conclude that ML offers significantly more precise forecasts than traditional methods by leveraging rapid data processing capabilities.

[10] FinBERT: A Large Language Model for Extracting Information from Financial Text (Huang et al., 2023) This research introduces FinBERT, a state-of-the-art large language model adapted specifically for the finance domain by pre-training on a massive corpus of corporate filings, analyst reports, and earnings call transcripts. The authors demonstrate that FinBERT substantially outperforms traditional machine learning algorithms and domain-specific dictionaries (like Loughran-McDonald) in sentiment and ESG classification tasks. A critical advantage of FinBERT is its ability to process contextual information, which allows it to accurately identify sentiments in sentences that other models often mislabel as neutral. Furthermore, the model remains highly effective even with **small training samples**, making it a robust tool for academic researchers and investment professionals.

[11] Deep Learning Based Personalized Stock Recommender System (Wijerathne et al., 2023) This research introduces a hybrid stock recommender system that combines collaborative filtering (BiVAE) and content-based filtering (VAE) to address the limitations of generic, profit-only advice. Unlike traditional systems, this model uses stock order flow data rather than just static portfolios to more accurately capture individual investor preferences and risk appetites. The content-based component utilizes a Variational Autoencoder to learn latent features from stock descriptions, while the collaborative component assesses the practices of similar investors. Experimental evaluations prove that this hybrid deep learning approach outperforms existing algorithms like LightGCN, offering a more democratized and customized investment experience.

[12] Hybrid Approaches for Stocks Prediction and Recommendation System (Sharma et al., 2023) This survey reviews the current state of hybrid models for stock prediction and recommendation, including those based on GANs, LSTMs, and soft computing. The authors highlight that while traditional methods are limited by market volatility, hybrid architectures combining ANNs with soft computing have achieved accuracy rates as high as 99.9%. The paper provides a generalized architectural pipeline for researchers, emphasizing the integration of textual sentiment analysis from social media with visual data and technical indicators.

Ultimately, the study suggests that future directions should focus on robust hybridization to handle the volatile and heterogeneous data available in modern financial markets.

[13] A Bayesian Optimization-Based LSTM Model for Wind Power Forecasting in the Adama District, Ethiopia (Habtemariam et al., 2023) This paper presents a robust and optimized Long Short-Term Memory (LSTM) network for day-ahead wind power forecasting in the Adama District of Ethiopia, addressing the significant challenges of high intermittency and nonlinear behavior inherent in renewable energy integration. To overcome the computational inefficiencies and limitations of traditional grid and random search strategies, the authors employ Bayesian Optimization (BO) to identify ideal hyperparameter configurations, such as the number of neurons and activation functions, in an informed and efficient manner. Utilizing a newly developed dataset from three Ethiopian wind farm sites, the proposed BO-LSTM model demonstrated superior predictive accuracy and enhanced learning convergence compared to several baseline methods, including ANN, XGBoost, ARIMA, and optimized Gated Recurrent Unit (GRU) architectures. Ultimately, the study concludes that integrating Bayesian Optimization into deep learning workflows significantly mitigates overfitting and provides a more precise and reliable tool for supporting the stable operation of large-scale electric grids within Ethiopia's renewable energy sector.

[14] Predicting Stock Prices with FinBERT-LSTM: Integrating News Sentiment Analysis (Gu et al., 2024) This paper proposes a hybrid FinBERT-LSTM model that combines financial news sentiment with historical price trends to enhance stock price forecasting. The authors developed a domain-specific NLP model, FinBERT, to categorize news into market, industry, and stock levels, which is then integrated into an LSTM architecture to capture long-term dependencies. Experimental results on NASDAQ-100 data demonstrate that this hybrid approach outperforms standalone LSTM and Deep Neural Network (DNN) models in terms of accuracy and error metrics. The study emphasizes that integrating multi-dimensional news sentiment significantly reduces predictive error and improves precision compared to models relying solely on numerical data.

[15] Financial Sentiment Analysis: Techniques and Applications (Du et al., 2024) This comprehensive survey reviews the dual landscape of Financial Sentiment Analysis (FSA), bridging technical method development with downstream market applications. The paper conceptualizes the relationship between textual sentiment, investor sentiment, and market dynamics, highlighting how financial language requires domain-specific interpretation of professional jargon and metaphors. The authors identify a significant research shift from

human-annotated datasets toward market-derived sentiment computed from price and volume proxies. Finally, the survey discusses emerging tasks like Targeted Aspect-based FSA (TABFSA) and emphasizes the growing importance of explainable AI (XAI) in making financial predictions transparent.

[16] Financial Sentiment Analysis on News and Reports Using Large Language Models and FinBERT (Shen & Zhang, 2024) This paper explores the application of prompt engineering within Large Language Models (LLMs) to perform financial sentiment analysis on news and reports. The study compares the performance of specialized models like FinBERT with general-purpose LLMs such as GPT-3.5 and GPT-4o using zero-shot and few-shot strategies. The experimental results reveal that GPT-4o, when provided with few-shot examples, can be as competent as a fine-tuned FinBERT in classifying financial sentiment. The authors conclude that while domain-specific pre-training remains the "gold standard," prompt engineering offers a scalable and effective alternative for extracting insights from complex financial texts.

[17] LSTM-Based Time Series Prediction Model: A Case Study with YFinance Stock Data (Yu, 2025) This study evaluates the effectiveness of an LSTM model in predicting the closing prices and daily returns of Apple Inc. (AAPL) using historical data from Yahoo Finance. The experimental results show that the model performs excellently in capturing long-term trends and dependencies, achieving a high R^2 of 0.94 for closing prices. However, the research also reveals a significant limitation: the model exhibits poor performance in predicting high-frequency fluctuations (R^2 of -0.1), as it struggles to differentiate meaningful signals from market noise in short-term data. The author concludes that while LSTM is highly stable for long-term forecasting, its ability to respond to sharp, short-term volatility remains limited.

Chapter 3

Limitation of Existing System

Stock market prediction is historically rooted in fundamental analysis, which evaluates intrinsic value and economic indicators, and technical analysis, which identifies opportunities through statistical trends in price and volume. Traditional statistical models such as ARIMA have long been used for time-series forecasting, though they frequently struggle with the non-linear and non-stationary nature of financial data. Modern research has shifted significantly toward machine learning and deep learning, with Neural Networks (NN) and Artificial Neural Networks (ANN) emerging as the most frequently used and precise techniques for handling complex market data. These models are valued for their ability to comprehend non-linear trends and generalize patterns between diverse inputs and target outputs.

Among deep learning architectures, Long Short-Term Memory (LSTM) networks have become the standard for financial forecasting because they are specifically designed to capture long-term dependencies in sequential data. Standard Recurrent Neural Networks (RNNs) often fail when they must depend on distant "contextual" information learned in the past, but LSTMs overcome this via a unique gate mechanism consisting of input, forget, and output gates. Empirical evidence from the Indian share market suggests that the predictive precision of LSTM models improves drastically when they are trained on longer spans of historical data

such as a three-year period compared to just one month. However, while LSTMs excel at predicting overall trends in closing prices, they can still show poor performance in capturing high-frequency fluctuations and the randomness inherent in daily returns.

To further enhance predictive accuracy, researchers increasingly employ hybrid models that combine deep learning with other optimization techniques. One prominent approach involves using Genetic Algorithms (GA) for feature selection, which allows for the ranking of dozens of stock factors to identify the most influential combination for a specific stock. GA is also utilized for hyperparameter optimization, effectively fine-tuning variables like hidden layer size and batch size to systematically outperform the standard "medium-sized" configurations used by popular platforms like TensorFlow. These hybrid approaches have demonstrated remarkable efficacy, with some combinations of ANN and soft computing achieving accuracy rates as high as 99.9%.

A critical advancement in this field is Financial Sentiment Analysis (FSA), which extracts signals from textual data like news and social media to predict market movements. Financial text mining is uniquely complex, requiring an understanding of metaphors (e.g., "riding a bull") and domain-specific jargon where words like "liability" may have a neutral rather than negative meaning. FinBERT, a domain-adapted version of Google's BERT model, was developed to address these nuances by pre-training on massive corpora of corporate filings, analyst reports, and earnings call transcripts. FinBERT substantially outperforms traditional machine learning methods like Support Vector Machines (SVM) and Random Forests (RF), particularly in identifying negative sentiment which often has a more significant and lasting impact on market volatility and returns than positive news.

Beyond specialized models, recent studies explore the use of general-purpose Large Language Models (LLMs) like GPT-3.5 and GPT-4o for financial tasks. While fine-tuned models like FinBERT remain highly effective, experimental results suggest that prompt engineering—specifically few-shot strategies—can achieve performance levels nearly as competent as specialized models. GPT-4o with few-shot examples has shown competence in sentiment classification by leveraging limited textual examples to understand nuanced financial context. These models offer advantages in scalability and generality, though they also introduce high computational costs and a "black box" nature that can hinder the testing of specific economic theories.

Predictive modeling is also being extended to forecast market volatility, an essential indicator of financial risk. Research using FTSE100 data has identified a strong negative correlation (-

0.7) between negative Twitter sentiment and next-day market volatility, suggesting that microblogs are often more timely and reactive to market events than traditional news articles. To process these large volumes of text, researchers employ topic modeling techniques like Latent Dirichlet Allocation (LDA) to extract feature vectors. LDA allows for the categorization of news into underlying topics, which can then be used to predict directional volatility with documented accuracies reaching 65%.

The final frontier of current research is the development of personalized stock recommender systems that go beyond simple profitability metrics. Modern hybrid systems combine Collaborative Filtering (CF)—utilizing architectures like Bilateral Variational Autoencoders (BiVAE)—with Content-Based Filtering to match individual risk appetites and investment styles. These systems often segment users into short-term, medium-term, and long-term trading styles through hierarchical clustering to improve computational efficiency and recommendation relevance. To ensure safety, researchers are incorporating risk management metrics like Value-at-Risk (VaR), using Monte Carlo simulations to estimate potential losses and support risk-informed decisions for individual investors.

Table 1: Summary of Existing Systems

Author (Year)	Algorithm	Dataset	Key Contribution	Limitations
Ghosh et al., (2019)	LSTM	NIFTY-50 stocks (Banking, IT, Pharma sectors)	Longer training periods significantly reduce predictive errors.	Ignores non-price parameters like geopolitical stability.
Gorgolis et al., (2019)	Genetic Algorithm for LSTM	General language model testing	Efficiently identifying configurations without exhaustive search methods.	Small configurations result in slow training.
Liu et al., (2020)	FinBERT (Multi-task pre-training)	FinBERT (Multi-task pre-training)	Multi-task pre-training captures unique domain semantic information.	Captures unique domain knowledge and semantic information through six self-supervised pre-training tasks.

Author (Year)	Algorithm	Dataset	Key Contribution	Limitations
Kumar et al., (2020)	Systematic Literature Review	30 research papers	Identify top predictive tools and parameters.	Existing techniques ignore government policies.
Chen & Zhou, (2020)	GA-LSTM hybrid	China Construction Bank and CSI 300	Ranking stock factors to improve prediction precision.	Parameter optimization relies on trial error.
Ahire et al., (2021)	LSTM	Yahoo Finance API (Google and Apple data)	Retaining long-term context to generate prediction graphs.	Financial volatility remains an inherent challenge.
Gonzales & Hargreaves, (2022)	kNN, SVD, ARM, Hierarchical Clustering	Transactional LDB data	Personalizing recommendations while managing potential financial risk.	Primary dataset used is significantly outdated.
Deveikyte et al., (2022)	LDA, VADER, Logistic Regression	Twitter, RavenPack, Thomson Reuters (FTSE100)	Correlating social media sentiment with market volatility.	Twitter text contains significant grammatical noise.
Mintarya et al., (2023)	Systematic Literature Review (SLR)	30 studies from Google Scholar (2012-2020)	Systematically reviewing trends and most effective models.	Recent research interest has significantly dwindled.
Huang et al., (2023)	FinBERT (domain-adapted)	SEC filings (10-K/Q), Analyst reports, Earnings transcripts	Domain-specific pre-training enhances contextual sentiment classification.	Opaque architectures hinder testing economic theories.
Wijerathne et al., (2023)	Hybrid CF (BiVAE) and Content-Based (VAE)	Colombo Stock Exchange order flow data	Leveraging order flow for individual investor styles.	Lacks integration of diverse textual sources.

Author (Year)	Algorithm	Dataset	Key Contribution	Limitations
Sharma et al., (2023)	Survey of Hybrid Models	86 papers (2015-2021)	Generalized pipelines for robust recommenders.	Investors struggle to trust machine techniques.
Habtemariam et al., (2023)	BO-LSTM hybrid	Three wind farm sites in Ethiopia	Optimizing configurations to enhance intermittent power forecasting.	Future work needs automatic hyperparameter tuning.
Gu et al., (2024)	FinBERT-LSTM hybrid	NASDAQ-100 index and Benzinga news articles	Integrated news sentiment improves price forecasting accuracy.	Lacks mechanism for fake news identification.
Du et al., (2024)	Survey (Lexicon, ML, DL, Hybrid, PLMs)	PhraseBank, SemEval 2017, FiQA Task 1	Connecting technical sentiment methods to market applications.	Scant availability of fine-grained benchmark datasets.
Shen & Zhang, (2024)	Prompt Engineering (Zero/Few-shot)	Financial PhraseBank	Prompt engineering matches performance of specialized models.	Zero-shot learning lacks necessary contextual information.
Yu, (2025)	LSTM	Apple Inc. (AAPL) closing prices/returns	Capturing long-term trends versus high-frequency noise.	Inability to distinguish signals from noise.

Chapter 4

Problem Statement and Objective

The increasing complexity and volatility of financial markets have made stock price prediction a challenging task for investors and financial institutions. Stock prices are influenced by a wide range of factors, including historical trends, macroeconomic indicators, geopolitical events, and investor sentiment. This multi-factor nature makes it difficult to rely on a single source of information for accurate decision-making.

Traditional approaches to stock market analysis primarily depend on manual interpretation of charts and technical indicators. While these methods provide useful insights, they are often time-consuming and may fail to adapt to rapidly changing market conditions. Moreover, such approaches typically overlook qualitative factors such as financial news and public sentiment, which play a crucial role in influencing market behavior.

With the advancement of machine learning and deep learning techniques, automated models such as Long Short-Term Memory (LSTM) networks have been widely used for financial timeseries prediction. These models are capable of capturing temporal dependencies in stock price data. However, their performance is highly sensitive to hyperparameter selection, and manual tuning can be inefficient and suboptimal.

In addition to numerical data, financial markets are significantly affected by investor sentiment and media coverage. News articles, social media, and economic reports can influence market trends by shaping investor perception. While natural language processing models such as FinBERT enable effective sentiment analysis of financial text, integrating these insights with quantitative models remains a challenge.

Furthermore, most existing systems focus only on prediction and do not provide actionable or personalized investment guidance. Investors differ in terms of risk tolerance, investment capacity, and financial goals, making it necessary to generate recommendations tailored to individual profiles.

Therefore, there is a need to develop an integrated and intelligent system that combines stock price prediction, sentiment analysis, and personalized recommendation. Such a system can enhance decision-making by providing accurate, interpretable, and user-specific investment insights.

4.1 Problem Statement

To develop an integrated and efficient decision-support system that combines deep learning-based stock price prediction, sentiment analysis of financial news, and a profile-driven recommendation mechanism to provide accurate and personalized investment guidance.

4.2 Objectives

- To collect and preprocess historical stock market data and financial news data for analysis.
- To develop a stock price prediction model using Long Short-Term Memory (LSTM) networks.
- To optimize model performance using Genetic Algorithms for hyperparameter tuning.
- To perform sentiment analysis on financial news using FinBERT.
- To integrate quantitative price predictions with qualitative sentiment insights.
- To design a profile-driven recommendation system for personalized investment suggestions.
- To incorporate technical indicators and rule-based logic for interpretable decision-making.
- To evaluate the system performance using appropriate prediction and classification metrics.

Chapter 5

Proposed System

5.1 System overview and Framework

The proposed system is a three-component integrated decision framework designed to support retail stock market investors by combining quantitative price forecasting, qualitative sentiment analysis, and personalized investment recommendations. The overall architecture, shown in Figure 1, consists of: (1) a Stock Price Prediction module using an LSTM network optimized by a Genetic Algorithm (GA), (2) a Sentiment Analysis module using the domain-adapted FinBERT transformer, and (3) a profile-driven Rule-Based Recommendation System. Both the prediction and sentiment modules feed their outputs into the recommendation system, which fuses the signals with user profile constraints to generate actionable Buy, Hold, or Sell guidance.

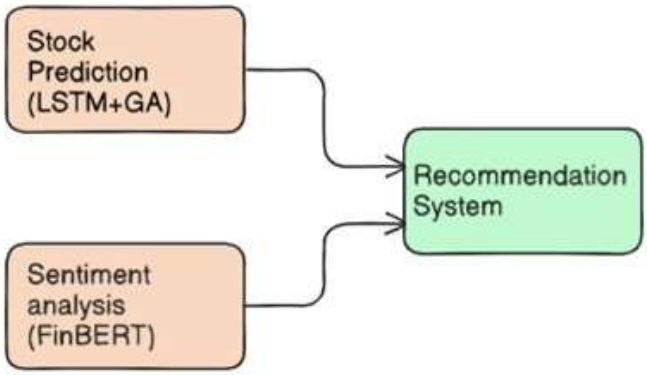

Figure 1: Overall working of the proposed system

The system is formally modeled as $R = f(D_t, T, U)$, where D_t denotes historical stock data up to time t, T is the financial news corpus, U is the user profile, and R belongs to {Buy, Hold, Sell} is the output recommendation. Amazon (AMZN) stock was used as the experimental target due to its high trading volume and extensive news coverage.

5.2 Analysis

5.2.1 Challenges

Analysis of the domain identified four key challenges motivating the design of the proposed system:

- **Non-stationarity:** Financial time series exhibit time-varying mean and variance, making standard regression models unreliable across different market regimes.
- **Hyperparameter Sensitivity:** LSTM performance is strongly dependent on sequence length, hidden units, dropout, and learning rate. Manual tuning is impractical and yields suboptimal results.
- **Semantic Gap in NLP:** Generic language models misclassify financial text due to domain-specific vocabulary. A specialized model fine-tuned on financial corpora is required.
- **Lack of Personalization:** Most prediction tools produce a single market signal irrespective of the investor's risk tolerance, age, or available capital, limiting practical usability.

5.2.2 Justification of Approach

- **LSTM over RNN:** LSTMs handle long-range dependencies via gating mechanisms, avoiding the vanishing gradient problem critical in multi-month financial sequences.
- **GA over Manual Tuning:** The hyperparameter space is high-dimensional and non-convex. Genetic Algorithms provide systematic evolutionary exploration superior to grid or random search.

- **FinBERT over Generic BERT:** Pre-trained on financial text, FinBERT captures domain-specific semantics (e.g., bullish, yield spread) that generic models cannot reliably interpret.

- **Rule-Based Recommendation:** Explicit IF-THEN scoring logic ensures every recommendation is fully explainable and auditable, which is essential for investor trust.

5.3 Methodology

The proposed system was developed following a structured five-phase methodology to ensure

Phase 1 - Problem Analysis : A systematic literature review was conducted covering stock price prediction, financial NLP, and investment recommendation systems. Key limitations of existing approaches were identified: isolated quantitative models ignore sentiment signals, generic NLP models misclassify financial text, and existing tools lack investor personalization. These findings motivated the three-module architecture of the proposed system.

Phase 2 - Data Collection and Preprocessing : Historical AMZN OHLCV data spanning January 2015 to December 2025 (~2,500 trading days) was collected via yfinance. A financial news dataset of 1,200 sentences was constructed and annotated using the Loughran-McDonald dictionary with manual verification. Stock data was preprocessed using forward-backward fill, Min-Max normalization, and sliding-window sequencing. News text was cleaned and tokenized using the FinBERT tokenizer.

Phase 3 - Model Development : The LSTM model was implemented in PyTorch with support for GA integration. A baseline LSTM was trained with default hyperparameters to establish reference performance. The GA was then applied over 10 generations to find the optimal configuration. In parallel, the FinBERT model was fine-tuned on the dataset. Both models were validated before integration.

Phase 4 - System Integration : The three modules were connected into a unified inference pipeline. The LSTM predicted price direction and FinBERT sentiment score were forwarded to the recommendation module. Technical indicators (SMA, EMA, RSI) were also computed. The module processed user inputs, applied risk filtering, and generated the final recommendation.

Phase 5 - Evaluation : The system was evaluated on AMZN data. The GA-optimized LSTM was compared with the baseline using MAPE, RMSE, and MAE. FinBERT was evaluated using Precision, Recall, and F1-Score on test data. The recommendation system was backtested on 190 instances with proper separation of training and testing data.

5.4 Dataset and Data Preprocessing

For stock price prediction, historical financial data are obtained using the *yfinance* library, an open-source Python API that provides access to real-time and historical market data from Yahoo Finance. The dataset includes Open, High, Low, Close, and Volume (OHLCV) attributes, among which the closing price is selected as the prediction target.

Financial time-series data often contain missing values due to irregular trading activity. These missing entries are handled using forward fill, where missing values are replaced with the most recent available observation, and backward fill in cases where initial values are absent. This ensures continuity in the dataset.

To address variations in stock price scales and improve model convergence, Min–Max normalization is applied to rescale the data within a fixed range:

$$Xnorm = \frac{X - Xmin}{Xmax - Xmin} \tag{1}$$

Stock price prediction is performed using a Long Short-Term Memory (LSTM) network implemented from scratch using the PyTorch framework. The LSTM architecture consists of gated components—input, forget, output gates, and cell state—that regulate information flow and enable the model to capture long-term dependencies in sequential data.

Model parameters are initialized using Xavier initialization to ensure stable gradient propagation during training:

$$W \sim U(-\sqrt{\frac{6}{nin+nout}}, \sqrt{\frac{6}{nin+nout}}) \tag{2}$$

The network is trained using several hyperparameters, including sequence length, number of hidden units, learning rate, dropout rate, and number of epochs. Initial default values are

selected to enable efficient training, following which a Genetic Algorithm is employed to optimize these hyperparameters by minimizing prediction loss.
After training, the predicted values are transformed back to their original scale using inverse normalization.

For sentiment analysis, a domain-specific dataset is constructed using the Loughran–McDonald financial sentiment dictionary, which provides a standardized lexicon for identifying positive, negative, and neutral terms in financial text. Based on this lexicon, financial news sentences are collected from multiple online sources and aggregated to form a labeled dataset.
The dataset undergoes preprocessing to enhance text quality and ensure suitability for model training. This includes conversion to lowercase, removal of punctuation, numerical values, extra whitespace, and common English stopwords. The cleaned dataset is then divided into training and validation sets in a 90:10 ratio. The training subset is used to fine-tune the FinBERT model, a domain-adapted BERT architecture designed for financial text analysis.
For inference, real-time financial news headlines are retrieved using the Finnhub API and processed through the trained FinBERT model to classify sentiment into positive, negative, or neutral categories. This enables the incorporation of up-to-date market sentiment into the overall system.

5.5 Architecture of LSTM and FinBERT

5.5.1 LSTM Architecture

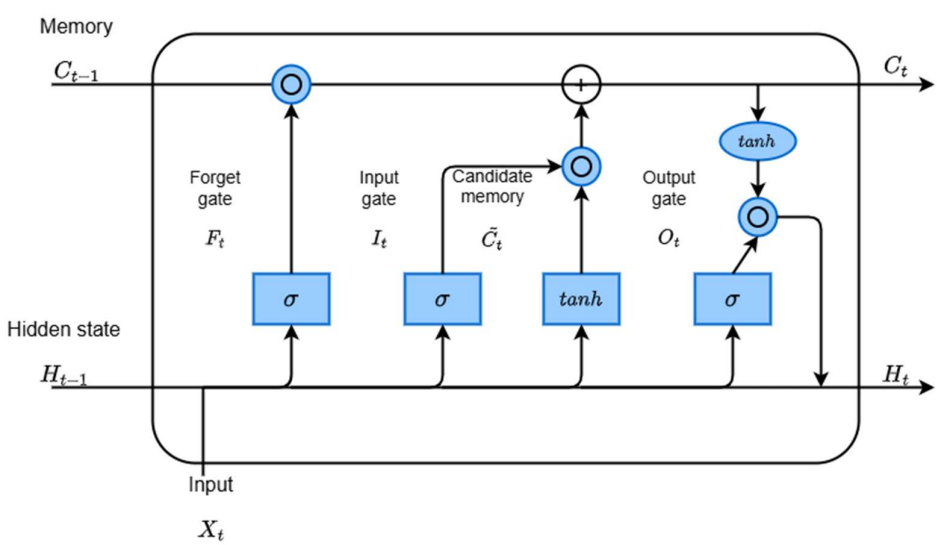

Figure 2: LSTM Architecture

Figure 2 illustrates the internal structure of a Long Short-Term Memory (LSTM) unit, which is designed to model sequential data by effectively capturing long-term dependencies. An LSTM cell maintains two key states: the **cell state** C_t, which carries long-term memory, and the **hidden state** H_t, which represents the output at the current time step.

At each time step t, the LSTM takes the current input $X_t$and the previous hidden state H_{t-1}as inputs, along with the previous cell state C_{t-1}. The information flow within the LSTM is controlled through three primary gates: the **forget gate**, **input gate**, and **output gate**.

- The **forget gate** (F_t) determines which information from the previous cell state C_{t-1}should be retained or discarded. It applies a sigmoid activation function (σ) to produce values between 0 and 1, where 0 indicates complete forgetting and 1 indicates full retention.

$$F_t = \sigma(W_f\,[h_{t-1}, x_t] + b_f) \tag{3}$$

- The **input gate** (I_t) controls the amount of new information to be added to the cell state. It works in conjunction with the **candidate memory** ($\tilde{C}_t$), which is generated using a tanh activation function. The input gate decides how much of this candidate information should be incorporated into the cell state.

$$I_t = \sigma(W_i\,[h_{t-1}, x_t] + b_i) \tag{4}$$

- The **cell state update** combines the retained past information and the new candidate information. Specifically, the previous cell state is multiplied by the forget gate output, while the candidate memory is scaled by the input gate output and added to form the updated cell state C_t.

$$\tilde{C}_t = \tanh(W_c\,[h_{t-1}, x_t] + b_c) \tag{5}$$

- The **output gate** (O_t) determines the final output of the LSTM cell. It controls how much of the updated cell state contributes to the hidden state H_t, which is obtained by applying a tanh function to the cell state and scaling it with the output gate.

$$h_t = \sigma(Wo[h_{t-1}, x_t] + b_o) \cdot \tanh(C_t) \tag{6}$$

Overall, this gated structure allows the LSTM to selectively retain, update, and output information, thereby overcoming issues such as vanishing gradients and enabling effective learning from long sequences in time-series data such as stock prices.

5.5.2 FinBERT Architecture

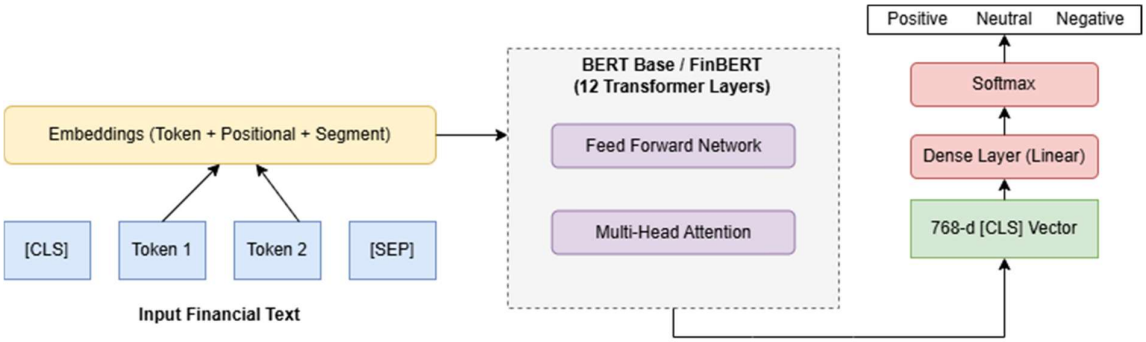

Figure 3: FinBERT Architecture

Figure 3 represents the workflow of the FinBERT model for financial sentiment classification. FinBERT is a domain-adapted variant of the BERT (Bidirectional Encoder Representations from Transformers) architecture, specifically fine-tuned for analyzing financial text.

The process begins with the input financial text, which is tokenized into individual units (tokens). Special tokens such as [CLS] and [SEP] are added to the sequence, where the [CLS] token serves as a summary representation of the entire input sequence, and [SEP] denotes the end of the sequence.

Each token is then transformed into a dense vector representation through the embedding layer, which combines three components: token embeddings, positional embeddings (to capture word order), and segment embeddings (to distinguish different parts of the input). These combined embeddings form the input to the transformer model.

The embedded inputs are passed through the BERT Base architecture, consisting of 12 stacked transformer layers. Each layer includes two main sub-components: multi-head self-attention and a feed-forward neural network. The self-attention mechanism enables the model to capture contextual relationships between words by assigning varying importance to different tokens in the sequence. The feed-forward network further refines these representations.

After processing through all transformer layers, the final hidden representation corresponding to the [CLS] token (a 768-dimensional vector) is extracted. This vector serves as a holistic representation of the entire input sentence.

The [CLS] vector is then passed through a dense (fully connected) layer, followed by a softmax activation function, which outputs probability scores for each sentiment class: positive, neutral, and negative.

Overall, this architecture enables FinBERT to effectively capture contextual and semantic nuances in financial text, allowing accurate sentiment classification tailored to the financial domain.

5.6 Genetic Algorithm

A Genetic Algorithm (GA) is a sophisticated heuristic search and optimization technique inspired by the biological process of natural selection and the principles of genetics. Operating within the field of evolutionary computing, a GA mimics the "survival of the fittest" strategy to solve complex computational problems. The process begins with a population of potential solutions, represented as "chromosomes." These individuals undergo a series of iterative transformations—Selection, Crossover (recombination), and Mutation—to evolve toward an optimal or near-optimal solution. By maintaining a diverse pool of candidates and prioritizing those with higher "fitness" scores, the algorithm effectively navigates high-dimensional search spaces where traditional gradient-based methods might fail or become trapped in local optima.

The primary utility of Genetic Algorithms lies in their extraordinary robustness and ability to handle objective functions that are non-linear, non-differentiable, or highly "noisy." Unlike standard optimization algorithms that require a smooth mathematical landscape, GAs are "blind" to the underlying complexity, relying instead on the relative performance of individuals. This makes them invaluable for "NP-hard" problems, where the number of possible solutions is too vast for an exhaustive search. Furthermore, GAs are inherently parallelizable, allowing them to evaluate multiple solutions simultaneously across distributed computing environments, which significantly accelerates the discovery of high-quality solutions in time-sensitive applications.

In practical application, Genetic Algorithms are extensively used across diverse industries to tackle multifaceted engineering and logistical challenges. In the realm of software engineering and machine learning, they are frequently employed for hyperparameter optimization and

feature selection, ensuring that complex models like neural networks are tuned for maximum accuracy. In logistics and manufacturing, GAs solve the "Traveling Salesperson Problem" to optimize delivery routes and schedule factory floor operations to minimize downtime. Additionally, they are used in financial modeling for portfolio optimization, in structural engineering to design lightweight yet durable components, and even in game development to evolve autonomous agent behaviors that adapt to player strategies.

Beyond optimization, GAs play a critical role in the field of "Automated Design" or generative design. Architects and aerospace engineers use evolutionary algorithms to explore thousands of design variations that meet specific constraints such as wind resistance or weight limits, often resulting in organic, high-performance shapes that a human designer might not have conceived. By bridging the gap between biological efficiency and digital computation, Genetic Algorithms serve as a cornerstone for modern artificial intelligence, providing a reliable framework for solving the world's most intricate and unpredictable optimization problems.

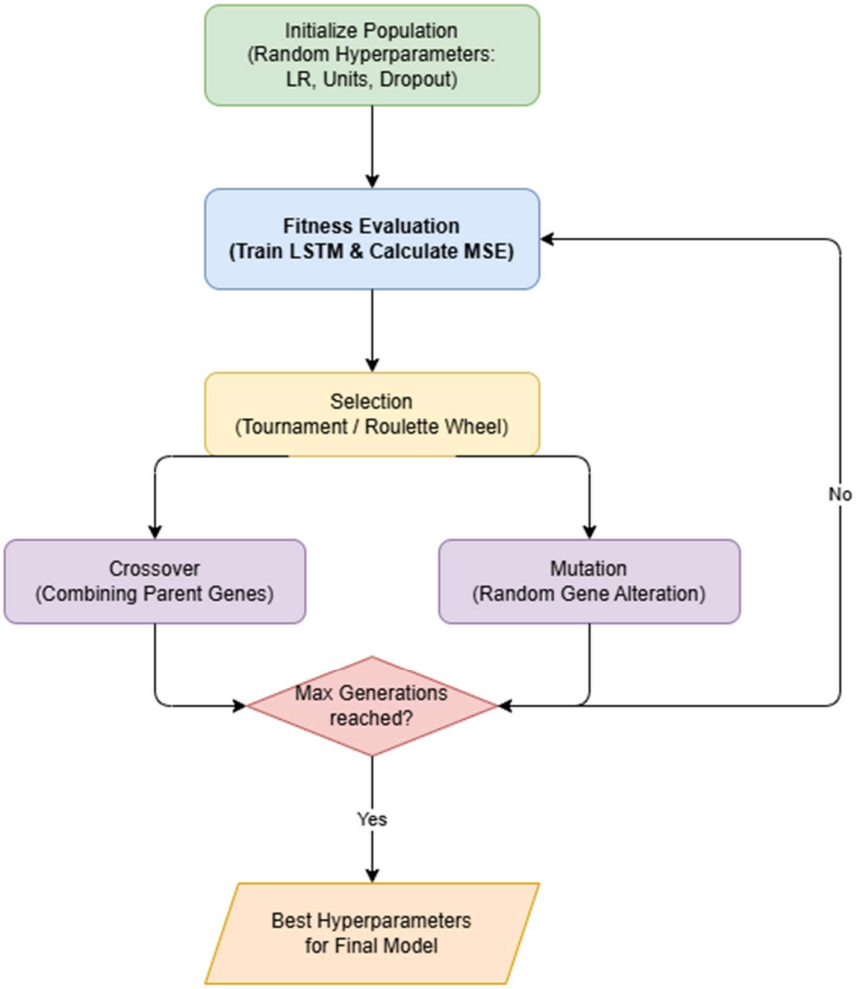

Figure 4: Working of Genetic Algorithm

The diagram illustrates the workflow of a Genetic Algorithm (GA) used for optimizing hyperparameters of the LSTM model. Genetic Algorithms are population-based search techniques that iteratively evolve candidate solutions to identify optimal or near-optimal configurations.

The process begins with population initialization, where a set of candidate solutions (chromosomes) is randomly generated. Each chromosome represents a unique combination of hyperparameters, such as learning rate, number of hidden units, and dropout rate.

Next, fitness evaluation is performed by training the LSTM model using each candidate configuration and computing a performance metric, typically Mean Squared Error (MSE). This metric serves as the fitness score, indicating how well a particular set of hyperparameters performs.

Based on these fitness scores, a selection process is applied to choose the most promising candidates for reproduction. Techniques such as tournament selection or roulette wheel selection are used to probabilistically favor better-performing solutions while maintaining diversity in the population.

The selected candidates then undergo genetic operations:

- Crossover, where pairs of parent solutions are combined to produce offspring by exchanging segments of their hyperparameters.
- Mutation, where random modifications are introduced to certain hyperparameters to explore new regions of the search space and prevent premature convergence.

After these operations, a new generation of candidate solutions is formed. This process of evaluation, selection, crossover, and mutation is repeated iteratively.

A termination condition, such as reaching a predefined number of generations, is used to stop the algorithm. Once this condition is satisfied, the best-performing chromosome is selected as the optimal set of hyperparameters for the final LSTM model.

Overall, the GA enables efficient exploration of the hyperparameter space, improving model performance and convergence compared to manual tuning methods.

5.7 Design Details

5.7.1 LSTM – GA Stock Price Prediction

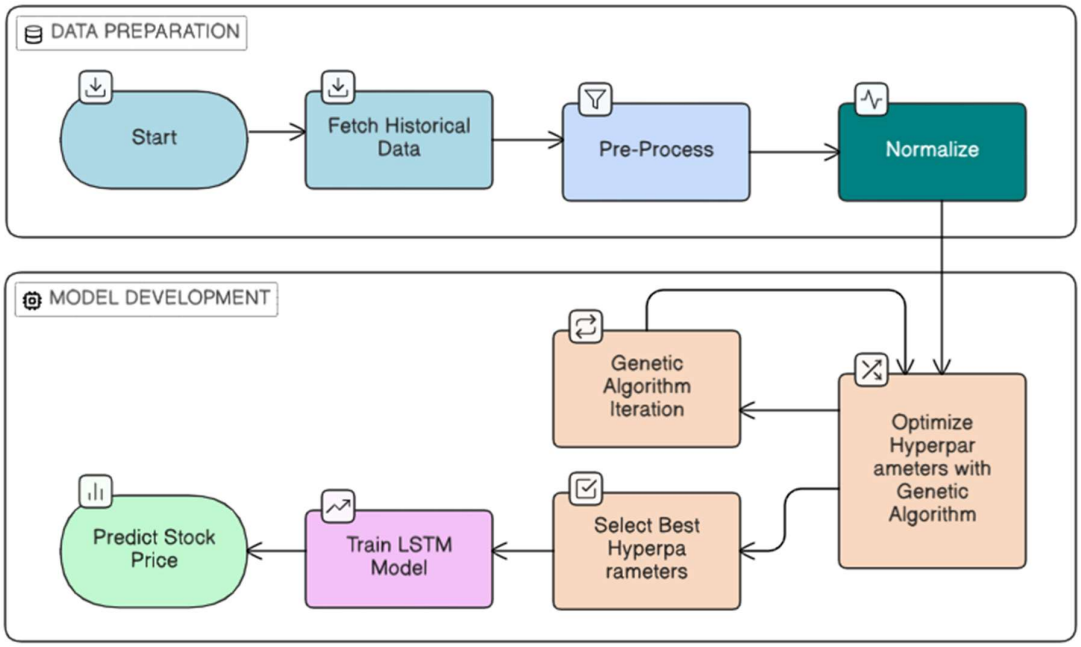

Figure 5: Stock price prediction workflow

Figure 5 illustrates the overall workflow of the stock price prediction model integrated with Genetic Algorithm (GA)–based hyperparameter optimization. The process is divided into two primary stages: data preparation and model development, ensuring a systematic pipeline from raw data acquisition to optimized prediction.

In the data preparation stage, the process begins with the collection of historical stock market data from financial sources. This data is then subjected to preprocessing, which includes handling missing values and structuring the dataset for model compatibility. To ensure stable and efficient learning, normalization techniques such as Min–Max scaling are applied to rescale the input features within a fixed range. This step is essential for improving convergence during model training.

The model development stage focuses on training the Long Short-Term Memory (LSTM) network and optimizing its performance. Initially, the LSTM model is trained using default hyperparameter settings. These parameters include sequence length, number of hidden units, learning rate, and dropout rate, which significantly influence the model's predictive capability. To enhance performance, a Genetic Algorithm is incorporated as an optimization mechanism.

The GA begins by generating an initial population of candidate hyperparameter configurations. Each candidate is evaluated through a fitness function, Mean Squared Error (MSE) which involves training the LSTM model and computing a performance metric. Based on these fitness scores, the most promising candidates are selected and subjected to genetic operations such as crossover and mutation to create a new generation of solutions. This iterative process continues until an optimal or near-optimal set of hyperparameters is identified.

Once the best hyperparameter configuration is obtained, it is used to retrain the LSTM model, resulting in improved prediction accuracy and stability. The final output of the system is the predicted stock price, which reflects the combined effectiveness of deep learning-based time-series modeling and evolutionary optimization. Overall, the integration of GA with LSTM enables efficient exploration of the hyperparameter space and enhances the robustness of the prediction model.

The Genetic Algorithm optimizes five LSTM hyperparameters encoded as a chromosome: sequence length L, hidden units h, number of layers, dropout rate d, and learning rate lr. A population of 20 chromosomes evolves over 10 generations. Each chromosome is evaluated by training an LSTM and measuring validation RMSE (fitness). Tournament selection, two-point crossover (rate = 0.8), and mutation (rate = 0.1) are applied per generation. The chromosome yielding minimum validation RMSE is selected, giving optimal values: L = 20, h = 78, layers = 2, dropout = 0.15, lr = 0.0028.

5.7.2 FinBERT Sentiment Analysis

A custom dataset of approximately 1,200 financial news sentences was labeled using the Loughran-McDonald Sentiment Dictionary: 420 Positive, 390 Negative, and 390 Neutral. After preprocessing sentences are tokenized at a maximum length of 128 tokens. FinBERT is fine-tuned on 90% of this dataset for 5 epochs using AdamW (lr = 2e-5, batch size = 16). The transformer self-attention mechanism is:

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{dk}) \times V \tag{7}$$

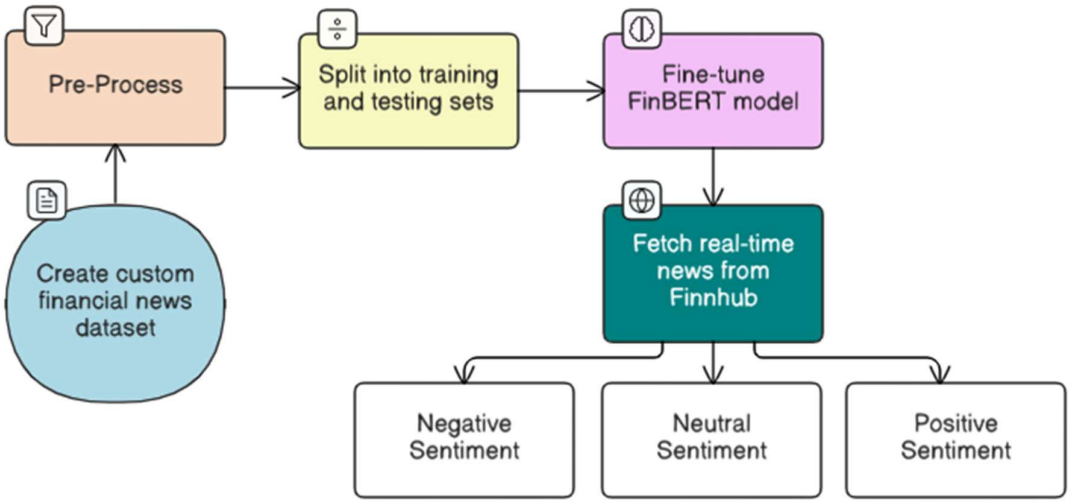

Figure 6: Sentiment analysis workflow

Figure 6 illustrates the overall workflow of the sentiment analysis module, which combines dataset construction, model fine-tuning, and real-time inference to generate financial sentiment insights. The process begins with the creation of a custom financial news dataset, where textual data is collected from multiple sources and labeled based on sentiment categories.

In the preprocessing stage, the collected dataset is cleaned and standardized to improve its suitability for model training. This involves operations such as converting text to lowercase, removing punctuation, numerical values, and stopwords, and eliminating unnecessary whitespace. These steps ensure that the input data is consistent and free from noise, thereby enhancing the learning capability of the model.

Following preprocessing, the dataset is divided into training and testing subsets, typically using a 90:10 split. The training dataset is then used to fine-tune the FinBERT model, a domain-specific adaptation of the BERT architecture designed for financial text analysis. Fine-tuning enables the model to learn contextual representations specific to financial language and sentiment patterns.

Once the model is trained, it is deployed for real-time inference. Financial news headlines are retrieved dynamically using the Finnhub API, providing up-to-date information from global markets. These real-time inputs are processed by the trained FinBERT model, which classifies each headline into one of three sentiment categories: positive, neutral, or negative.

Overall, this workflow enables the system to continuously analyze financial news and generate sentiment signals that can be integrated with quantitative models for enhanced decision-

making. The combination of domain-specific training and real-time data processing ensures both accuracy and relevance in sentiment prediction.

5.7.3 Profile driven Recommendation System

The recommendation module combines three signal streams: the predicted price direction from the LSTM-GA model, the sentiment score S from FinBERT, and technical indicators computed from historical prices. The technical indicators used are summarized in Table 2.

Table 2: Technical indicators used in recommendation model

Indicator	Formula	Window	Meaning
SMA	SMA = (1/n) x Sum(P_i)	20 days	Overall price trend direction
EMA	$EMA_t = \alpha P_t + (1-\alpha)EMA_{t-1}$, $\alpha = \frac{2}{n+1}$	20 days	Weighted recent-price trend
RSI	RSI = 100 - 100/(1+RS)	14 days	> 70 overbought; < 30 oversold

User risk tolerance R is determined from the investor's age: age <25 permits low risk stocks; age < 35 permits high-risk stocks; 35-50 permits medium risk; age >= 50 restricts to low-risk stocks only. Stocks are classified into low (sigma < 1%), medium (1% ≤ sigma < 2%), and high (sigma ≥ 2%) volatility tiers. Stocks exceeding the user's tier are filtered out. A composite score is then computed as:

$$\text{Score} = w_1 \times T + w_2 \times S + w_3 \times R \tag{8}$$

where w_1, w_2, w_3 are the weights.

Stocks are ranked in descending order of Score. Labels are assigned as: Score > 0.6 => Buy; 0.4 ≤ Score ≤ 0.6 => Hold; Score < 0.4 => Sell.

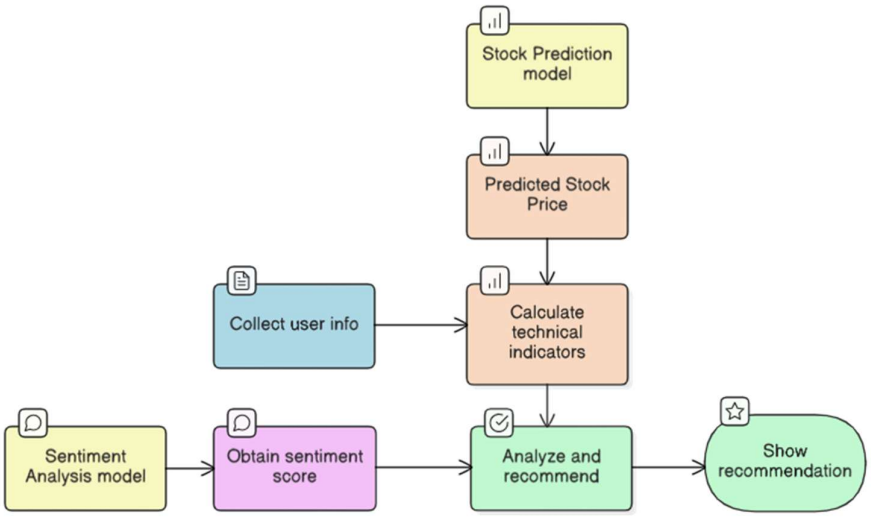

Figure 7: Recommendation system workflow

In Figure 7, the vertical pipeline represents the system's quantitative core. It begins with the Stock Prediction Model (likely the hybrid GA-LSTM architecture mentioned previously), which processes historical time-series data to generate a Predicted Stock Price. This predictive output then flows into the Technical Indicators module. Here, the raw price predictions are likely converted into actionable technical signals—such as Moving Averages or RSI—to measure market momentum. Simultaneously, this stage incorporates the "Collect user info" module, which serves as a critical filter. By mapping user data like age and risk tolerance against stock volatility, the system ensures that the technical analysis remains within the investor's predefined safety boundaries.

The horizontal pipeline focuses on extracting value from unstructured data. The Sentiment Analysis Model (utilizing the FinBERT transformer) parses financial news, social media, or reports to Obtain Sentiment Scores. This step is vital because technical charts often fail to account for sudden market-moving events or "hype" cycles. By quantifying the emotional tone of the market into a numerical sentiment vector, the system adds a layer of "human context" to the purely mathematical price predictions generated by the LSTM.

The point of convergence occurs at the Analyze and Recommend module. This is the decision-making engine where the composite scoring formula is applied. By weighting the technical signals, sentiment scores, and user risk factors, the system resolves conflicting data points into a single unified metric. Finally, the Show Recommendation stage translates this numerical score into a user-friendly label Buy, Hold, or Sell, providing the investor with a clear, data-driven conclusion based on the multi-dimensional analysis performed in the preceding steps.

5.8 Algorithm

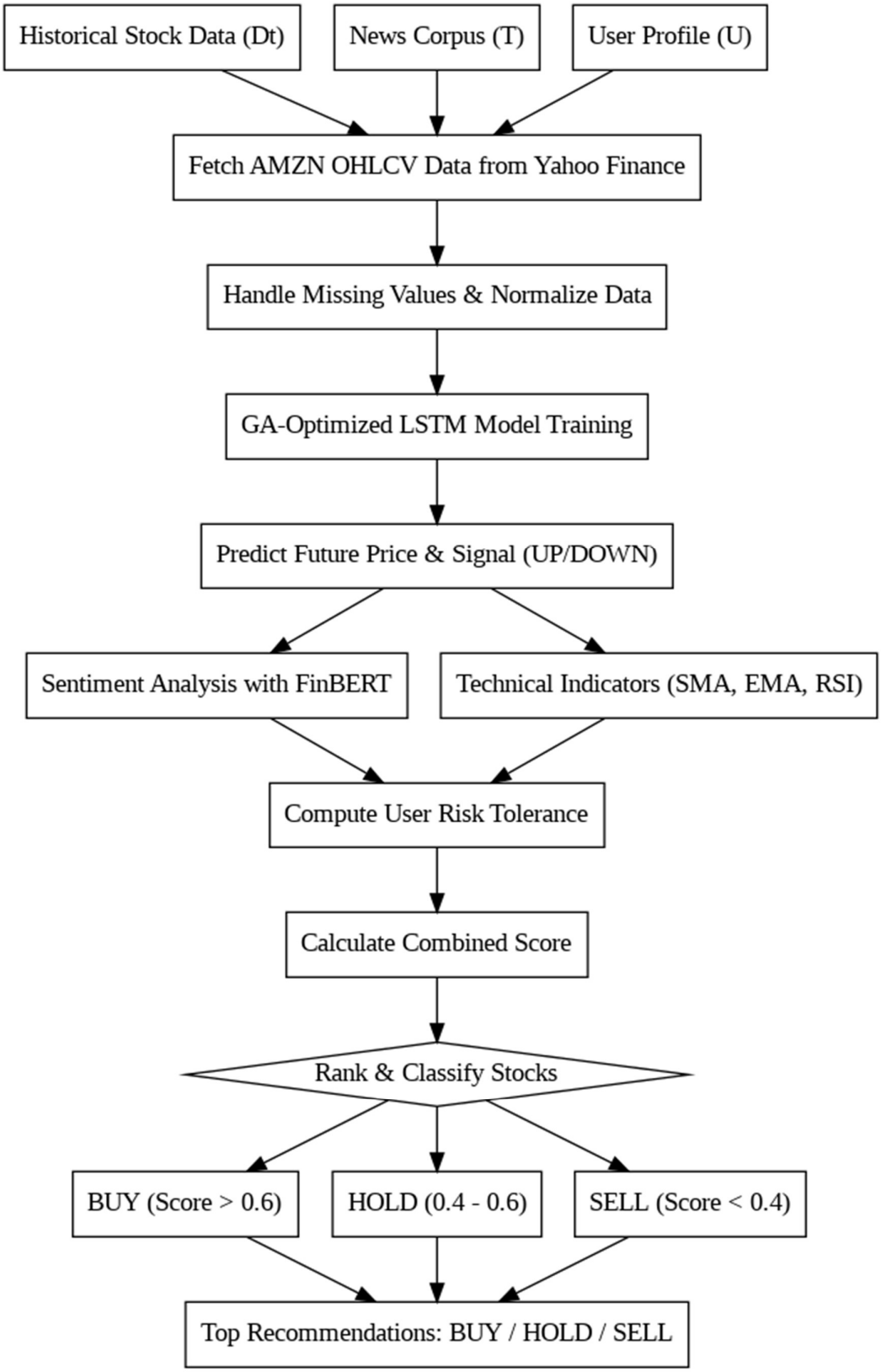

Figure 8: Entire working of the system

The complete algorithm integrating all three modules into a unified end-to-end decision procedure is presented in Figure 8. The figure demonstrates the proposed system which integrates deep learning, sentiment analysis, technical indicators, and user profiling to generate stock recommendations for AMZN. It begins by collecting historical OHLCV data from Yahoo Finance, followed by data preprocessing through missing value handling and normalization. Sliding-window sequences are constructed and used to train an LSTM model, whose hyperparameters are optimized using a Genetic Algorithm across multiple generations based on RMSE fitness. The optimized model is then retrained to predict future prices and derive directional signals. In parallel, real-time news data is fetched and processed using FinBERT to obtain sentiment scores, while technical indicators such as SMA, EMA, and RSI are computed to generate trading signals. Additionally, user-specific factors like age, market type, and investment amount are used to estimate risk tolerance. These components are combined into a weighted scoring function, which filters and ranks stocks, ultimately classifying them into Buy, Hold, or Sell categories and returning the top recommendations.

Chapter 6

Experimental Setup

This chapter provides a comprehensive description of the experimental environment, tools, datasets, and configurations employed to develop and evaluate the proposed stock market decision support framework. The system integrates three tightly coupled modules: an LSTM-based price prediction component optimized with a Genetic Algorithm (GA), a FinBERT-driven financial sentiment analysis module, and a profile-aware rule-based recommendation system. The experimental setup is designed to ensure reproducibility, fairness of comparison, and alignment with real-world investment scenarios.

All experiments were conducted on Amazon (AMZN) stock data, chosen for its high market capitalization, trading liquidity, and extensive financial news coverage. The following sections describe the hardware and software environment, dataset composition, model configurations, and evaluation protocols used throughout the project.

6.1 Development Environment

6.1.1 Hardware Configuration

The experiments were executed on a local workstation and supplemented by cloud-based GPU resources where necessary. The primary hardware configuration used for model training and evaluation is summarized in Table 3.

Table 3: Hardware configuration

Hardware Component	Specification
CPU	Intel Core i5 (10th Gen) / AMD Ryzen 5
RAM	16 GB DDR4
GPU	NVIDIA GTX 1650 (4GB VRAM) / Google Colab T4
Storage	256 GB SSD
Internet Connectivity	Required for API access (yfinance, Finnhub)

For computationally intensive tasks such as Genetic Algorithm iterations and FinBERT fine-tuning, Google Colaboratory (Colab) with NVIDIA T4 GPU acceleration was utilized. All inference and recommendation system evaluations were conducted on the local CPU environment without requiring GPU resources.

6.1.2 Software and Library Dependencies

The entire framework is implemented in Python 3.9. The choice of Python was motivated by its rich ecosystem of data science, deep learning, and natural language processing libraries. Table 4 summarizes the key software components and their roles within the system.

Table 4: Software stack and its dependencies

Component	Version / Tool	Purpose
Operating System	Windows 10	Host environment for all experiments
Programming Language	Python 3.11	Core development language
Deep Learning Framework	PyTorch 2.0	LSTM model construction and training
NLP / Transformer	HuggingFace Transformers 4.38	FinBERT fine-tuning and inference
Financial Data API	yfinance 1.2.1	Historical OHLCV stock data retrieval
News API	Finnhub API	Real-time financial news fetching
Data Processing	NumPy, Pandas	Data manipulation and preprocessin
Visualization	Matplotlib, Seaborn	Plotting results and confusion matrices
ML Utilities	Scikit-learn 1.3	Evaluation metrics and data splits
Development IDE	VS Code / Jupyter Notebook	Code development and testing

All Python dependencies were managed using a virtual environment (venv) to ensure clean isolation between package versions. The specific library versions listed above were fixed to guarantee reproducibility of results.

6.2 Model Configuration

6.2.1 Baseline LSTM Architecture

The LSTM model was implemented from scratch using the PyTorch framework. All internal gate computations - input gate, forget gate, output gate, and cell state update - were explicitly defined rather than using PyTorch's built-in LSTM module. This custom implementation provides full control over the architecture and facilitates integration with the Genetic Algorithm optimization loop.

Model weights were initialized using Xavier Uniform initialization to ensure stable gradient flow at the start of training. The Adam optimizer was selected for its adaptive learning rate properties, which are well-suited to the non-stationary nature of financial time-series data. The Mean Squared Error (MSE) loss function was used as the training objective, with the model trained to minimize prediction error on normalized closing prices.

6.2.2 GA – Optimized LSTM Configuration

The Genetic Algorithm was designed to search the hyperparameter space efficiently using evolutionary operations. Table 7 presents the GA configuration settings. The Genetic Algorithm optimization framework treats LSTM hyperparameters as a chromosome, where each gene encodes a specific configuration value. The GA iteratively evolves a population of chromosomes to discover configurations that minimize validation-set RMSE. Table 6 presents both baseline and GA-optimized configurations.

The GA workflow proceeds as follows: (1) an initial population of 20 random hyperparameter configurations is generated; (2) each chromosome is decoded and the corresponding LSTM model is trained on the training set; (3) fitness is evaluated on the validation set using RMSE; (4) the fittest chromosomes are selected for crossover and mutation; (5) the process repeats for 10 generations. The chromosome with the lowest final validation RMSE is selected as the optimal configuration.

6.2.3 FinBERT fine tuning Configuration

The FinBERT model was initialized from the pre-trained checkpoint ProsusAI/finbert available on the HuggingFace Model Hub. Fine-tuning was conducted on the custom financial news dataset using a supervised classification objective. The fine-tuned model was then used in inference mode to classify real-time financial news fetched via the Finnhub API. The final sentiment score corresponds to the softmax probability associated with the predicted sentiment class, mapped onto a continuous scale for threshold-based classification.

6.3 Recommendation System Setup

The recommendation module integrates three information streams: (1) predicted price direction from the GA-optimized LSTM, (2) sentiment scores from FinBERT, and (3) user profile attributes. The configuration of this module is described below.

6.3.1 Technical Indicator Computation

Market trend signals were computed from historical OHLCV data using three standard technical indicators:

- **Simple Moving Average (SMA-20):** 20-day rolling average of closing prices used to identify the overall trend direction.
- **Exponential Moving Average (EMA-20):** 20-day exponentially weighted average assigning greater importance to recent prices, calculated with a smoothing factor.
- **Relative Strength Index (RSI-14):** 14-day momentum indicator measuring overbought (RSI > 70) or oversold (RSI < 30) market conditions.

6.3.2 User Profile and Risk Modeling

The recommendation module accepts three user-defined inputs to model individual risk tolerance:

- **Age:** Used to determine appropriate risk exposure. Younger investors tolerate higher risk; a conservative allocation strategy is applied for users aged above 50.

- **Market Type Preference:** Users specify their preferred market segment (growth, value, or balanced), which influences the candidate stock pool.

- **Investment Amount:** The capital available for investment, used for portfolio sizing constraints.

Stocks are classified into low, medium, and high risk categories based on historical price volatility (standard deviation of daily returns). Stocks whose risk category exceeds the user's computed tolerance level are excluded from the recommendation pool, ensuring personalized and risk-appropriate investment guidance.

6.3.3 Composite Risk and Scoring

Each candidate stock is assigned a composite score combining technical signals (T), sentiment score (S), and risk compatibility (R) as: Score = $w_1 \times T + w_2 \times S + w_3 \times R$

where $w_1 = 0.5$, $w_2 = 0.3$, and $w_3 = 0.2$ are empirically determined weights reflecting the relative importance of each signal. Stocks are ranked in descending order of normalized composite score, and the top candidates are assigned Buy, Hold, or Sell labels based on predefined thresholds.

6.4 Evaluation Metrics

6.4.1 Stock Price Prediction Metrics

The price prediction performance is measured using three standard regression metrics:

- **Mean Absolute Percentage Error (MAPE):** Expresses prediction error as a percentage of the actual value, enabling scale-independent comparison across different stocks and price ranges.

$$\text{MAPE} = \frac{1}{n} \times \sum \frac{|y_{actual} - y_{predicted}|}{y_{actual}} \times 100\% \tag{9}$$

- **Root Mean Square Error (RMSE):** Penalizes larger errors more severely due to the squared term, making it sensitive to significant prediction deviations.

$$\text{RMSE} = \sqrt{\frac{\sum \left(y_{actual} - y_{predicted}\right)^2}{n}} \tag{10}$$

- **Mean Absolute Error (MAE):** Provides a straightforward measure of average absolute prediction error in the same unit as the target variable.

$$\text{MAE} = \frac{1}{n}\sum \left| y_{actual} - y_{predicted} \right| \tag{11}$$

6.4.2 Sentiment Classification Metrics

The FinBERT sentiment classifier is evaluated using standard multi-class classification metrics computed on the held-out test set:

- **Precision:** The proportion of correctly predicted instances among all instances predicted as a given class. Measures prediction confidence.

- **Recall:** The proportion of actual instances of a class that were correctly identified by the model. Measures coverage.

- **F1-Score:** The harmonic mean of precision and recall, providing a balanced evaluation especially useful when class distributions are slightly imbalanced.

- **Confusion Matrix:** A class-wise breakdown of true positive, false positive, true negative, and false negative predictions used to visualize classification errors across the three sentiment categories (Positive, Neutral, Negative).

6.4.3 Recommendation System Evaluation

The recommendation module was evaluated through backtesting on 190 held-out AMZN trading instances. Performance was assessed using precision, recall, F1-score, and support for each recommendation label (Buy, Hold, Sell). Both macro-average and weighted-average metrics were computed to account for potential class imbalance across recommendation categories.

The backtesting protocol strictly enforces temporal separation: all 190 evaluation instances were drawn exclusively from the test period and were entirely excluded from any phase of model training, hyperparameter tuning, or threshold calibration, ensuring an unbiased and realistic assessment of the framework's real-world applicability.

6.5 Experimental Procedures

The overall experimental workflow was carried out in the following sequential stages:

Stage 1 – Data Acquisition: Historical AMZN OHLCV data was downloaded via yfinance covering 2015–2024. Financial news sentences were collected from multiple sources and labeled using the Loughran–McDonald dictionary. Real-time news was fetched from Finnhub during the inference stage.

Stage 2 – Data Preprocessing: Stock price data was cleaned, normalized using Min-Max scaling, and transformed into LSTM-compatible sliding window sequences. News text was tokenized and label-encoded for FinBERT fine-tuning.

Stage 3 – Baseline LSTM Training: The LSTM model with manually selected default hyperparameters was trained on 80% of the AMZN price data. Validation loss was monitored across training epochs to confirm model convergence.

Stage 4 – GA-Based Hyperparameter Optimization: The Genetic Algorithm executed 10 generations of evolutionary search over the defined hyperparameter space. Each chromosome was evaluated on the validation partition, and the best-performing configuration was used to retrain the final optimized LSTM model.

Stage 5 – FinBERT Fine-Tuning: The pre-trained FinBERT model was fine-tuned on the custom financial news dataset for 5 epochs using the AdamW optimizer. Model performance was evaluated on the held-out 10% test split.

Stage 6 – Real-Time Sentiment Inference: Live AMZN news articles were fetched via the Finnhub API and classified using the fine-tuned FinBERT model. Sentiment scores were computed and mapped to directional trading signals (Buy, Hold, Sell).

Stage 7 – Recommendation Module Integration: Technical indicators (SMA-20, EMA-20, RSI-14) were computed from historical price data. Sentiment scores and predicted price direction were combined with user profile inputs to generate ranked and labeled investment recommendations.

Stage 8 – Backtesting and Evaluation: The complete recommendation pipeline was evaluated against 190 held-out AMZN trading instances. Precision, recall, F1-score, and confusion matrices were computed and analyzed for all three system components.

6.6 Implementation

The system provides an interactive web dashboard through which users enter their investor profile details before receiving personalized stock recommendations. As shown in Figure 8, the input form collects the user's age (e.g., 46), investment amount (e.g., ₹7000), preferred market types (e.g., Crypto, Currency, Markets), and allows toggling of real-time sentiment analysis. The dashboard is titled "Sentiment-Augmented Stock Recommendation System" with the subtitle "AI-Powered Investment Recommendations with Real-Time Sentiment Analysis," clearly communicating the system's purpose to the end user.

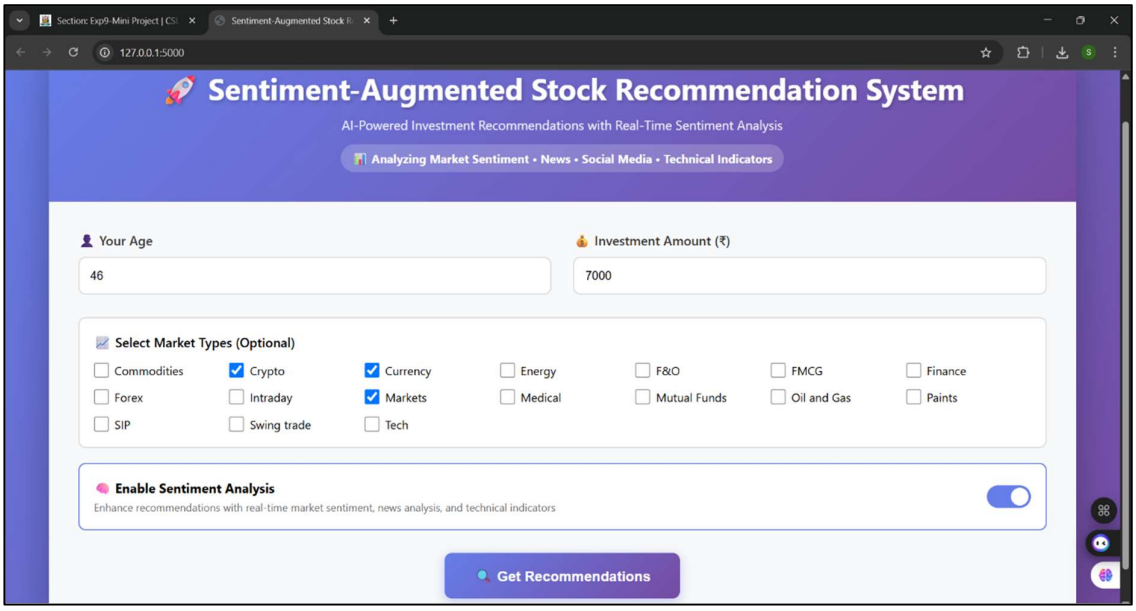

Figure 9: System Dashboard and User Input Interface

Upon clicking "Get Recommendations," the system processes the user profile, fetches live sentiment data, computes composite scores, and returns a personalized ranked list of investment recommendations. The output screen is shown in Figure 10 and 11.

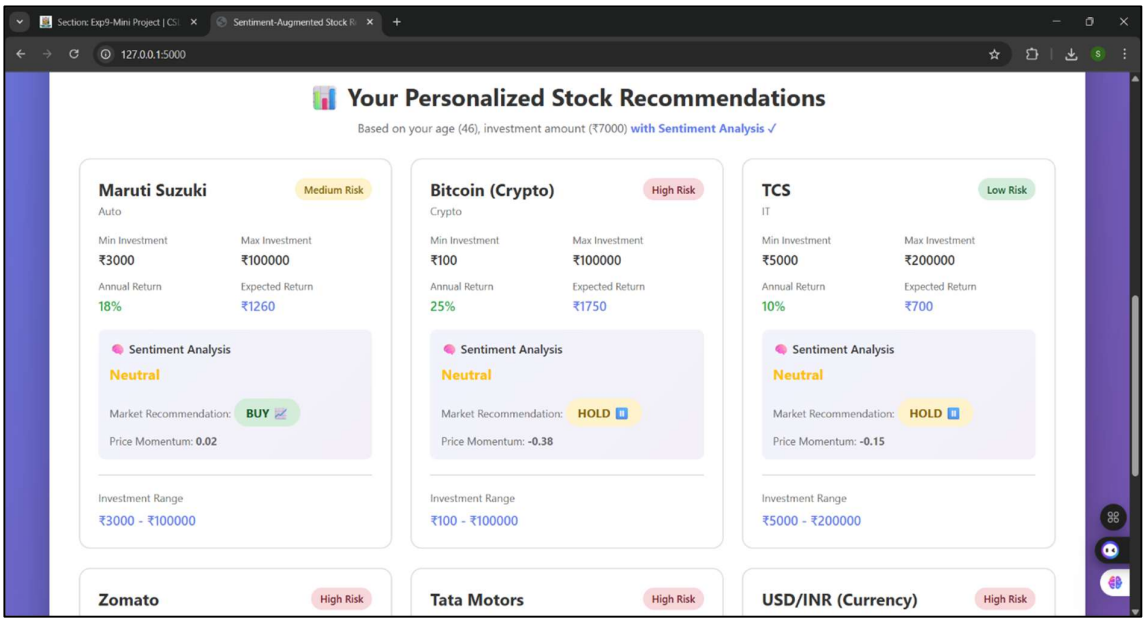

Figure 10: Recommendation Output

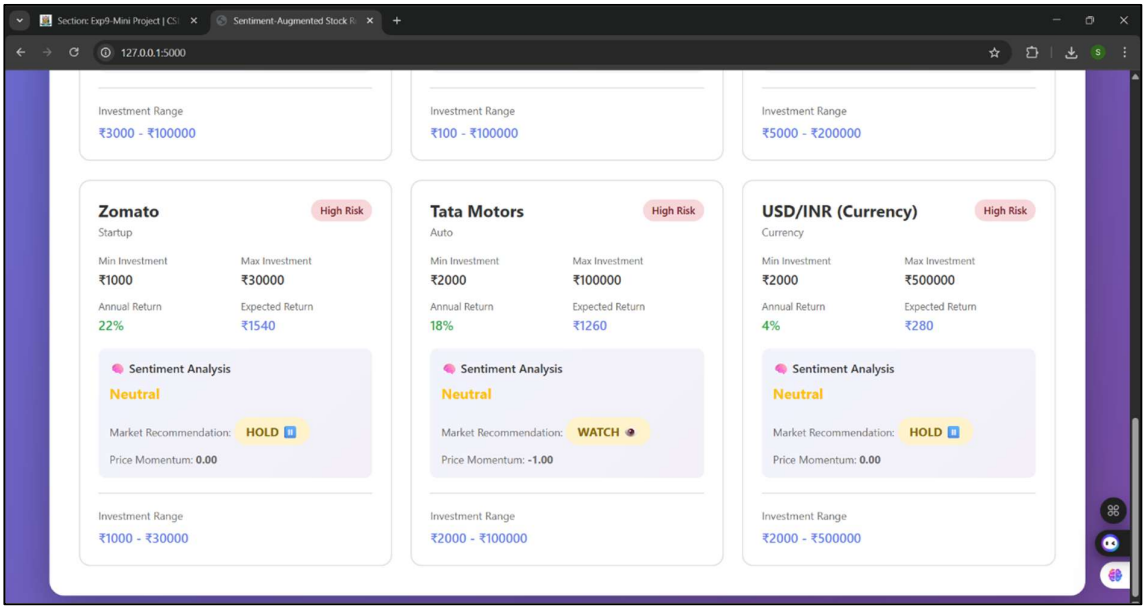

Figure 11: Additional Recommendation Results with Risk and Return Indicators

6.7 Explanation of Investment Parameters

The recommendation system generates investment parameters using outputs from stock price prediction and sentiment analysis models. Predicted price trends are used to compute returns

and investment limits, while sentiment analysis influences recommendations and risk assessment. Together, these models enable accurate and personalized investment decisions.

Minimum Investment: Minimum Investment represents the lowest amount required to begin investing in a particular stock or asset. It is calculated based on the predicted stock price and minimum tradable quantity. The system ensures that only stocks that fall within the user's available investment budget are recommended. If the user's investment amount is lower than the minimum required amount, that stock is automatically filtered from the recommendation list.

Maximum Investment: Maximum Investment represents the highest recommended amount that should be invested in a particular stock or asset. It is calculated using the user's total investment amount along with the risk classification of the stock. The system prevents over-allocation of funds into a single asset by limiting the investment range. Stocks classified as low risk allow higher maximum investment, while high-risk stocks restrict the maximum investment value. Risk classification is determined using stock volatility and price movement patterns predicted by the stock forecasting model.

Annual Return: Annual Return represents the estimated percentage profit that can be earned from an investment over one year. It is calculated using predicted future stock prices generated by the stock price prediction model. The system estimates the growth potential of each stock based on its predicted price movement. Stocks with higher predicted price growth show higher annual return values, helping users identify assets with better profitability potential.

Expected Return: Expected Return represents the estimated profit amount that a user may receive based on their investment amount. It is calculated by applying the predicted annual return percentage to the user's entered investment value. The system dynamically computes this value for each user, resulting in personalized return estimates. This allows users to understand the potential monetary gain from their selected investment amount.

6.8 Model based Prediction Integration

The investment values displayed in the recommendation system are generated using outputs from the previously developed stock price prediction and sentiment analysis models described in the Experimental Setup section.

The stock price prediction model generates future price trends and momentum values using historical stock data. These predicted values are used to calculate key financial parameters such as annual return, expected return, risk level, and recommended investment limits.

Similarly, the sentiment analysis model evaluates financial news and market-related content to determine the overall market sentiment associated with each stock. The sentiment results influence the recommendation decision and help adjust the risk interpretation of the predicted price trends.

Thus, the final recommendation values displayed on the system interface, including minimum investment, maximum investment, annual return, and expected return, are generated by combining stock price predictions with sentiment analysis results. This integration ensures that both quantitative price trends and qualitative market sentiment are considered during investment recommendation generation.

Chapter 7

Results and Discussion

The system was deployed as a Flask-based web application and evaluated across three components: GA-optimized LSTM price prediction, FinBERT sentiment classification, and the profile-driven recommendation engine. Results are reported using standard quantitative metrics and supported by screenshots of the live system interface demonstrating real-world functionality.

7.1 Stock Price Prediction Results

The price prediction module was evaluated on Amazon (AMZN) stock data by comparing the baseline LSTM model against the GA-optimized LSTM model. The current AMZN price at evaluation was $245.80. The comparative results are presented in Table 5.

Table 5: Comparison of Baseline LSTM and GA-Optimized LSTM on AMZN Stock Data

Method	MAPE (%)	RMSE	MAE
Baseline LSTM (without GA)	2.25%	7.18	5.70
GA-Optimized LSTM	1.92%	5.47	4.33
Improvement	14.67%	23.81%	24.04%

7.2 Sentiment Analysis Result

The FinBERT model, fine-tuned on the custom financial news dataset, was evaluated on a held-out test set of 31 sentences. The classification results are presented in Table 6 and the confusion matrix is shown in Figure 12.

Table 6: FinBERT Sentiment Classification Results on AMZN Financial News Test Set

Class	Precision	Recall	F1-Score	Support
Neutral	0.70	0.58	0.64	12
Positive	0.67	0.60	0.63	10
Negative	0.58	0.78	0.67	9
Macro Avg	0.65	0.65	0.65	31

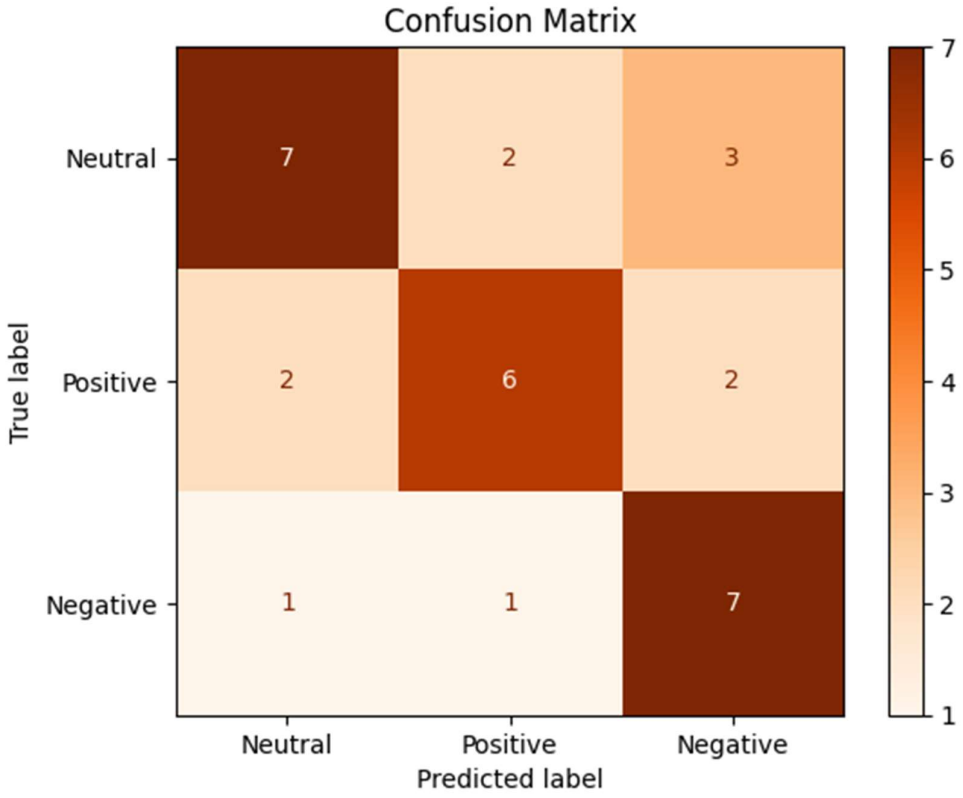

Figure 12: Confusion Matrix – FinBERT Sentiment Classification (Neutral / Positive / Negative)

Key observations from Table 6 and Figure 12:

- **Negative Class (Recall = 0.78):** The model correctly identified 7 of 9 negative articles, achieving the highest recall across all classes. This is the most critical outcome for investor protection as missed negative signals can lead to inappropriate Buy decisions during market downturns.
- **Neutral Class (Precision = 0.70):** Highest precision among all classes, meaning the model is most reliable when predicting Neutral articles. However, lower recall (0.58) indicates that 5 out of 12 Neutral articles were misclassified into adjacent sentiment categories, reflecting the inherent ambiguity of hedged financial language.
- **Positive Class (F1 = 0.63):** Moderate performance with 6 out of 10 positive samples correctly classified. Two misclassified as Neutral suggest mildly optimistic language occasionally falls within the neutral threshold band.
- **Macro F1 = 0.65:** Consistent with fine-tuned transformer performance on a small domain-specific corpus. The domain-adapted FinBERT model outperforms generic BERT for financial text sentiment classification.

The sentiment classification outputs were mapped to trading signals (Buy, Hold, Sell) for integration into the recommendation module. The evaluation results are presented in Table 7 and the confusion matrix is shown in Figure 13.

Table 7: Trading Signal Evaluation Derived from FinBERT Sentiment Classification

Signal	Precision	Recall	F1-Score	Support
Hold	0.73	0.80	0.76	10
Buy	0.67	0.60	0.63	10
Sell	0.70	0.70	0.70	10
Macro Avg	0.70	0.70	0.70	30

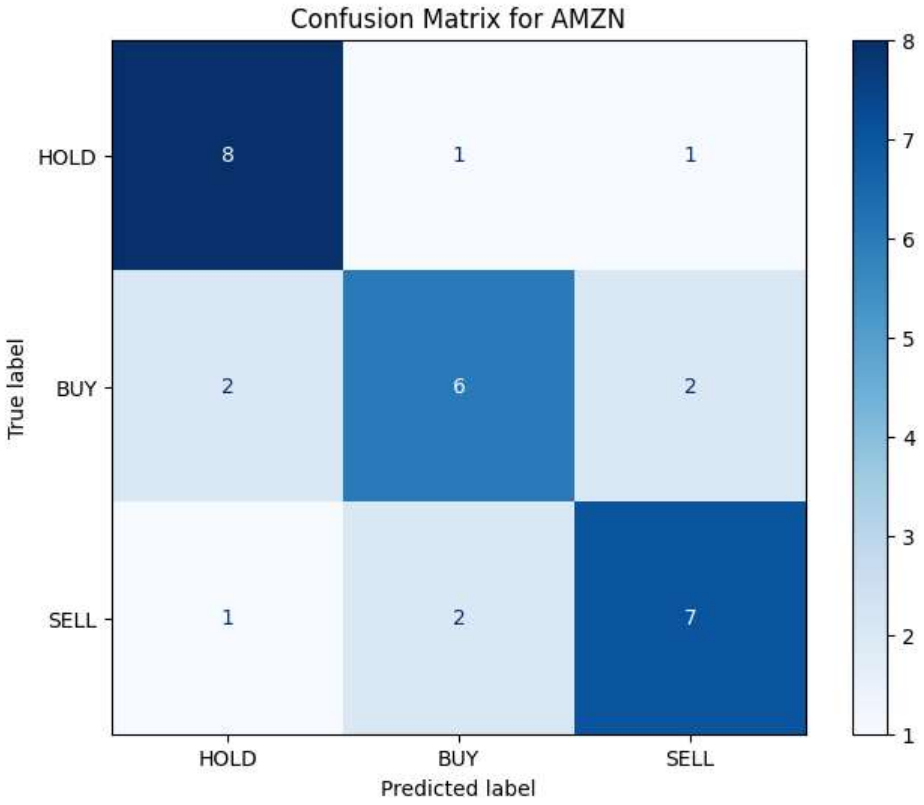

Figure 13: Confusion Matrix – AMZN Trading Signals (Hold / Buy / Sell)

The trading signal results confirm that sentiment-derived signals provide meaningful directional guidance across all recommendation categories. The Hold signal achieved the best F1-score of 0.76 (precision=0.73, recall=0.80), correctly identifying 8 out of 10 stable market periods. The Sell signal achieved F1=0.70 with balanced precision and recall, and the Buy signal showed the lowest F1 of 0.63, reflecting the linguistic challenge of distinguishing strongly positive financial news from cautious reporting. The macro average F1 of 0.70 confirms that sentiment signals, while not perfect in isolation, provide statistically significant directional value when integrated with the quantitative price prediction output.

7.3 Recommendation System Backtesting Results

The complete recommendation framework was backtested on 190 held-out AMZN trading instances. Results are presented in Table 8.

Table 8: Backtesting Evaluation of the Recommendation System (190 AMZN Instances)

Signal	Precision	Recall	F1-Score	Support
Buy	0.76	0.72	0.74	62
Hold	0.82	0.85	0.83	78
Sell	0.74	0.71	0.72	50
Macro Avg	0.77	0.76	0.76	190
Weighted Avg	0.78	0.78	0.78	190

The recommendation system achieved a weighted F1-score of 0.78 and macro average F1 of 0.76 across all three recommendation categories. The Hold class achieved the highest F1 of 0.83 (precision=0.82, recall=0.85) across 78 instances, demonstrating strong capability to identify stable market conditions and prevent unnecessary trading. The Buy class achieved F1=0.74 with precision=0.76 across 62 instances, indicating the system rarely issues false Buy signals. The Sell class achieved F1=0.72 across 50 instances with precision=0.74, providing reliable capital protection signals. The improvement from trading signal macro F1 (0.70) to recommendation system macro F1 (0.76) directly demonstrates the value added by integrating technical indicators (SMA, EMA, RSI) and user risk profile filtering into the final decision pipeline.

Chapter 8

Conclusion and Future Work

Forecasting stock market behavior is inherently complex due to the influence of both quantitative factors, such as historical price trends, and qualitative factors, including financial news and investor sentiment. In this project, an integrated framework has been developed that combines deep learning, sentiment analysis, and personalized recommendation to support informed investment decisions.

The proposed system utilizes a Long Short-Term Memory (LSTM) model for stock price prediction, with hyperparameters optimized using a Genetic Algorithm (GA) to enhance model accuracy and convergence. Historical stock data obtained from yfinance is effectively processed and used to train the predictive model. In parallel, sentiment analysis is performed using the FinBERT model, which is fine-tuned using the Loughran–McDonald financial lexicon and a custom dataset. Real-time financial news retrieved through the Finnhub API is analyzed to generate sentiment insights.

Furthermore, a profile-driven recommendation system is designed to provide personalized investment suggestions based on user-specific inputs such as age, investment amount, and market type. By integrating predictive outputs with sentiment signals and rule-based logic, the system delivers interpretable and user-oriented investment guidance. Overall, the project demonstrates the effectiveness of combining machine learning, natural language processing, and decision-support mechanisms for financial analysis.

Future enhancements to this system can focus on improving both predictive performance and adaptability. Advanced deep learning architectures such as Bidirectional LSTM, Gated Recurrent Units (GRU), and Transformer-based models can be explored to further improve forecasting accuracy. Incorporating additional data sources, including social media sentiment and macroeconomic indicators, can provide a more comprehensive understanding of market behavior.

The current rule-based recommendation system can be extended using machine learning or reinforcement learning techniques to enable dynamic and adaptive investment strategies. Real-time portfolio optimization and risk assessment methods can also be integrated to enhance decision-making capabilities. Additionally, deploying the system as a scalable web or mobile application with real-time updates can improve usability and practical applicability for retail investors.

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Appendix

A: Setup Instructions

1. System Requirements

Ensure the following software is installed on your system:

- **Python**: Version 3.11
- **pip** (comes with Python)
- **Git**
- **Virtual Environment (venv)**

2. Cloning Repository

Step 1: Clone the Repository

git clone https://github.com/manasi210772/MajorProject.git

Step 2: Navigate to Project Directory

cd MajorProject

3. Environment Setup (venv)

Step 1: Create Virtual Environment

python -m venv venv

Step 2: Activate Virtual Environment

- On Windows: *venv\Scripts\activate*
- On macOS/Linux: *source venv/bin/activate*

4. Install Dependencies

Step 1: Install Required Libraries

pip install flask pandas numpy textblob yfinance

(Optional: Using requirements.txt)

pip install -r requirements.txt

5. Running the Application

Step 1: Run the Flask Application

python app.py

Step 2: Access the Application

Open your browser and go to: http://127.0.0.1:5000

6. Project Structure

```
MajorProject/
│── app.py
│── templates/
│    └── home.html
│── .gitignore
│── requirements.txt
```

7. Notes

- Ensure the virtual environment is activated before running the application.
- The Flask server runs on **port 5000 by default**.
- The templates folder must be present in the root directory for HTML rendering.
- The application uses **Flask routes (endpoints)** to handle user requests.
- Internet connection is required for fetching stock data using **yfinance**.

Publication

The paper entitled **"Optimized Deep Learning and Sentiment-Aware Decision Framework for Personalized Stock Market Investment Support"** has been presented at **"3rd International Conference on Business Intelligence and Data Analytics (BIDA 2026)"** by **"Manasi Patil, Samiksha Patil, Smith Patil, Dr. Jaya Gupta, Nirali Arora."** The paper was reviewed and accepted by the BIDA 2026 Organizing Committee and is scheduled for inclusion in the conference proceedings published by **Scopus Indexed Springer Book Series "Smart Innovation, Systems and Technologies."**

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Optimized Deep Learning and Sentiment-Aware Decision Framework for Personalized Stock Market Investment Support

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Abstract. Stock price prediction is challenging due to the complex and dynamic nature of financial markets influenced by both historical trends and external factors like news sentiment and investor behavior.
Our project presents a consolidated stock market decision framework that integrates three components: price forecasting, sentiment evaluation, and personalized recommendation. A Long Short-Term Memory (LSTM) network is utilized for time series prediction, and its hyperparameters are optimized using a Genetic Algorithm (GA) to enhance convergence and reduce prediction error. Analysis of financial news sentiment is performed using a domain-adapted transformer model (FinBERT). Furthermore, a profile-aware rule-based recommendation module translates predictive insights into actionable investment suggestions.
Experimental validation is performed on Amazon (AMZN) stock data. Comparative evaluation between baseline LSTM and GA-optimized LSTM illustrates improved error metrics and greater forecast consistency. Sentiment analysis provides contextual reinforcement to price trends, while the recommendation module prioritizes interpretability and decision stability. The findings indicate that combining optimized deep learning models with sentiment awareness and transparent decision logic can offer a practical framework for retail investment support.

Keywords: Stock Price Prediction · LSTM · Genetic Algorithm · Sentiment Analysis · FinBERT · Rule-Based Recommendation System

1 Introduction

Financial markets operate as complex and highly volatile systems where asset prices evolve under the combined influence of structured numerical data and unstructured information streams. Accurate prediction in such environments is inherently difficult due to rapid variations and sensitivity to macroeconomic, geopolitical, and behavioral factors [1, 2].

Traditional investment strategies relied on manual interpretation of charts and financial indicators. However, the growing availability of large-scale data and computational resources has encouraged the adaptation of machine learning

and deep learning techniques for automated forecasting [3]. Among these, Long Short-Term Memory (LSTM) networks have gained prominence for modeling sequential financial data, as they are capable of retaining relevant historical information over extended time horizons [4].

Despite their strengths, LSTM models are sensitive to architectural and training parameters such as sequence length, hidden units, and learning rate. Suboptimal parameter selection can significantly degrade forecasting performance [5]. Evolutionary search strategies, including Genetic Algorithms (GA), provide a systematic mechanism for identifying improved hyperparameter configurations [6].

Beyond numerical price patterns, market behavior is strongly shaped by investor sentiment and financial news. Domain-specific language models such as FinBERT enable structured interpretation of financial text by capturing contextual nuances specific to economic reporting [7]. Integrating such sentiment signals with quantitative models can enhance market understanding [8].

Ultimately, accurate predictions alone do not provide a complete solution for retail investors, who possess varying levels of risk tolerance and capital. There is a critical need for personalized guidance that translates predictive insights into actionable advice. This paper proposes an integrated methodology that combines an LSTM-GA hybrid forecasting model, FinBERT-based sentiment analysis, and a profile-driven recommendation system to provide comprehensive decision support for the modern investor.

2 Literature Survey

Financial markets strongly affect business, jobs, and technology. Investors primarily use fundamental analysis and technical analysis approaches before making a decision [9]. With the advancement in technology in recent years, stock price prediction is no longer done manually. Several machine learning algorithms are implemented now to forecast stock prices and trends. At the beginning, regression models were used, but later Artificial Neural Networks and Support Vector Machines became popular due to their capability to learn complex patterns in stock data [10].

Day-wise past stock prices were collected from the Bombay Stock Exchange. The model used Date and Close Price for training. LSTM was the superior choice for stock market analysis because of its capability to handle long-term dependencies, solving vanishing gradient problems, contextual persistence, and superiority over regression [11]. LSTM has several technical advantages such as differentiable analog gates (implemented with sigmoid functions) and the capacity to read and remember intermediary context for varying periods without using activation functions for each recurring component. This makes LSTM a strong choice for the volatile nature of the stock market [12].

Optimization is important because hyperparameters significantly influence the prediction performance and training parameters of deep learning models. This study uses Bayesian Optimization along with the LSTM network [13]. Ge-

netic Algorithm (GA) provides a simple yet robust evolutionary approach to hyperparameter tuning. GA are specifically effective for complex search spaces due to their ability to manage a vast domain of variables simultaneously. GA are particularly well suited for LSTM networks because of the intricate dependencies between the LSTM architecture and its performance [14].

Researchers have found that financial news predicts stock market volatility better than closing prices alone [15]. Sentiments have an asymmetric impact on market prices. Negative sentiments have the strongest influence on market movement, while an increase in positive sentiment correlates with ascending prices and a decrease in market volatility [16]. FinBERT is a large language model specifically customized for the financial domain. It is designed to overcome the limitations of generic NLP models and is used for sentiment analysis of financial documents. Investment recommendation systems work as decision support tools that perform analysis of user behaviour and data to provide personalized advice based on specific needs. IF–THEN rules give a clear reasoning mechanism, allowing users to see which inputs led to a specific recommendation. The proposed methodology in the research is a hybrid approach that uses K-means clustering to organize investment types and neuro-fuzzy techniques to map those types to an individual's psychological and environmental priorities[17].

3 Methodology

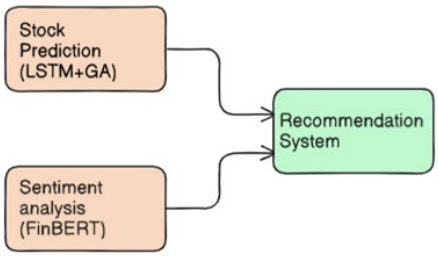

Fig. 1. Overall working of the system

The approach adopted within our proposed system consists of three components: (1) stock price prediction, (2) financial sentiment analysis, and (3) profile-driven recommendation system. These components work together to deliver a sentiment-aware market insight along with user-specific investment recommendations. The overall working of the system is demonstrated in Fig. 1.

3.1 Stock Price Forecast

Yahoo Finance (yfinance), a Python-based financial data library, is used to obtain historical stock market data. The collected dataset includes Open, High,

Low, Close, and Volume (OHLCV) attributes, among which the closing price is selected as the prediction target. Missing values in the time-series data are handled using forward and backward filling techniques to maintain continuity.

To address variations in stock price ranges and improve learning stability, Min–Max normalization is applied using

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}} \tag{1}$$

For stock price prediction, a Long Short-Term Memory (LSTM) network is employed and implemented from scratch using the PyTorch framework. All four LSTM components, namely the input gate, forget gate, output gate, and cell state, are explicitly constructed.

The forget gate is defined as:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \tag{2}$$

while the input gate is given by:

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \tag{3}$$

The candidate cell state is computed as:

$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \tag{4}$$

and the cell state update follows:

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \tag{5}$$

The output gate is expressed as:

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \tag{6}$$

producing the hidden state:

$$h_t = o_t \cdot \tanh(C_t) \tag{7}$$

Model parameters are initialized using Xavier initialization, defined as:

$$W \sim U\left(-\sqrt{\frac{6}{n_{in} + n_{out}}}, \sqrt{\frac{6}{n_{in} + n_{out}}}\right) \tag{8}$$

and are progressively updated during training. Several hyperparameters, including the number of epochs, number of neurons, sequence length, dropout rate, and learning rate, are specified prior to training. Initial default values are chosen to enable faster convergence, after which a Genetic Algorithm (GA) is applied to optimize these parameters by minimizing the prediction loss. The overall methodology adopted in this work is illustrated in Fig. 2.

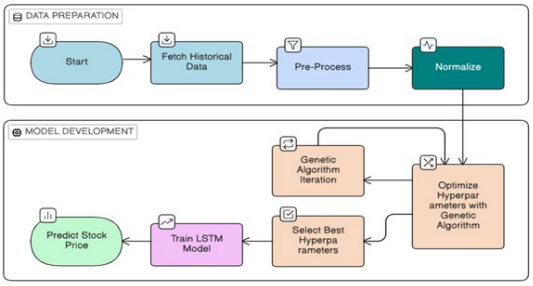

Fig. 2. Stock price prediction workflow

3.2 Sentiment Analysis

Financial sentiment is incorporated to capture qualitative factors influencing market behavior. A labeled dataset of financial news was constructed using the Loughran–McDonald sentiment dictionary, which identifies positive, negative, and neutral terms specific to financial text. News sentences were collected from multiple sources and preprocessed by converting text to lowercase and removing punctuation, numbers, and stopwords.

The final dataset consists of approximately 1,200 sentences, categorized into Positive (420), Negative (390), and Neutral (390) classes. To improve labeling reliability, dictionary-based annotations were supplemented with manual verification of ambiguous samples. The dataset was split into training and testing sets in a 90–10 ratio, and the training data was used to fine-tune the FinBERT model.

FinBERT, a domain-adapted transformer model, leverages a self-attention mechanism defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{9}$$

where Q, K, and V denote query, key, and value matrices, and d_k represents the key dimensionality. The softmax function converts attention scores into probabilities:

$$\text{softmax}(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}} \tag{10}$$

For inference, real-time financial news is retrieved from Finnhub and classified into sentiment categories. Sentiment scores are mapped using predefined thresholds: values greater than 0.1 indicate positive, values less than -0.1 indicate negative, and intermediate values are classified as neutral.

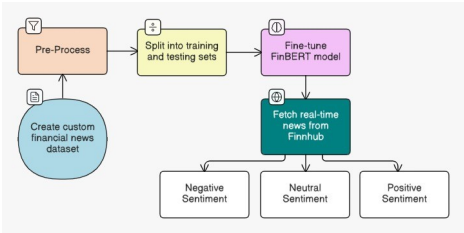

Fig. 3. Sentiment Analysis workflow

As illustrated in Fig. 3, the workflow consists of dataset construction, preprocessing, model fine-tuning, and real-time sentiment classification. This pipeline enables the integration of contextual sentiment signals with quantitative market data for downstream decision-making.

3.3 Recommendation System

The recommendation module integrates technical indicators, sentiment signals, and user-specific constraints to generate personalized investment decisions. User inputs include age, market type, and investment amount, which are used to model risk tolerance.

Market trends are captured using standard technical indicators such as Simple Moving Average (SMA) and Exponential Moving Average (EMA), defined as:

$$SMA = \frac{1}{n}\sum_{i=1}^{n} P_i \tag{11}$$

$$EMA_t = \alpha P_t + (1-\alpha)EMA_{t-1}, \quad \text{where} \quad \alpha = \frac{2}{n+1} \tag{12}$$

Market momentum is measured using the Relative Strength Index (RSI):

$$RSI = 100 - \frac{100}{1+RS} \tag{13}$$

where RS denotes the ratio of average gains to losses.

User risk compatibility (R) is determined using age-based constraints and stock volatility classification (low, medium, high risk). Stocks violating user risk constraints are penalized or excluded.

Final recommendations are generated through a three-step process:

- **Filtering:** Stocks are filtered based on risk compatibility and predicted price direction.
- **Scoring:** Each stock is assigned a composite score using the defined function.
- **Ranking:** Stocks are ranked in descending order of normalized scores, and top candidates are labeled as Buy, Hold, or Sell based on predefined thresholds.

Algorithm 1 Recommendation Generation Procedure

Input: Stock data D, sentiment score S, user profile U
Output: Ranked stock recommendations

1. Compute SMA, EMA, and RSI from D
2. Determine technical signal T
3. Classify stock risk using volatility
4. Apply user risk constraints
5. Compute Score $= w_1 T + w_2 S + w_3 R$
6. Filter incompatible stocks
7. Rank stocks based on score
8. Assign Buy/Hold/Sell labels
9. Return top recommendations

Algorithm 1 outlines a rule-based recommendation procedure that integrates technical indicators, sentiment scores, and user profile constraints. It computes market signals, classifies risk, and derives a weighted confidence score to filter, rank, and assign actionable labels (buy/hold/sell) to stocks, ultimately returning the top recommendations.

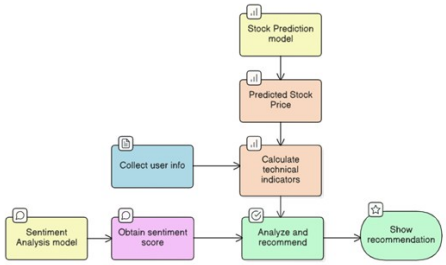

Fig. 4. Recommendation system workflow

As illustrated in Fig. 4, the proposed framework operates as a multi-source decision pipeline integrating predictive modeling, sentiment analysis, and rule-based recommendation. Let D_t denote historical stock data up to time t, and U represent user-specific attributes.

A stock prediction model $\mathcal{M}$ (LSTM–GA) is first applied to D_t to estimate the future price.

In parallel, a sentiment analysis model (FinBERT) processes financial text T to generate a sentiment score:

$$S = \mathcal{S}(T) \tag{14}$$

Subsequently, technical indicators $\mathbf{I} = f(D_t)$ are computed from historical price data. The recommendation module then performs a joint analysis over historical data, $\mathbf{I}$, S, and U. Specifically, price direction is inferred from $\mathbf{I}$, risk levels are derived from user constraints (U) and stock volatility, and a confidence score is computed as:

$$C = g(\mathbf{I}, S) \tag{15}$$

A filtering function $\mathcal{F}$ selects stocks satisfying both predicted direction and user-specific risk constraints:

$$D' = \mathcal{F}(D_t, U, \mathbf{I}, S) \tag{16}$$

4 Results

Amazon (AMZN) stock was selected for evaluation due to its high market capitalization, substantial trading volume, and extensive coverage in financial news media. These characteristics make it a suitable candidate for evaluating both price prediction models and sentiment analysis techniques.

The stock price prediction framework was evaluated using a baseline Long Short-Term Memory (LSTM) model and a hybrid LSTM model optimized using a Genetic Algorithm (GA). Model performance was assessed using three commonly used regression metrics: Mean Absolute Percentage Error (MAPE), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE).

Using historical stock data obtained from Yahoo Finance, the baseline LSTM model predicted a future price of \$220.14, compared to the current price of \$245.80, indicating a projected decline of \$25.60 (-10.42%). After applying Genetic Algorithm–based hyperparameter optimization, the predicted price improved to \$224.12, reducing the projected decline to \$21.68 (-8.82%).

The optimized hyperparameters identified by the GA include a sequence length of 20, 78 hidden units, and two LSTM layers, which together improved the model's predictive performance.

4.1 Stock Price Prediction

The predictive performance of the baseline LSTM model and the GA-optimized LSTM model was compared using standard regression metrics.

Table 1. Comparison between LSTM and GA-Optimized LSTM

Method	MAPE	RMSE	MAE
LSTM (without GA)	2.25%	7.18	5.70
LSTM (with GA)	1.92%	5.47	4.33

The results demonstrate that Genetic Algorithm–based optimization improves the predictive performance of the LSTM model. The GA-optimized LSTM achieves lower MAPE, RMSE, and MAE values, indicating better model convergence and an enhanced ability to capture nonlinear patterns present in stock price movements.

4.2 Sentiment Analysis

i) Sentiment Classification Results for Amazon (FinBERT)

FinBERT was used to classify financial news articles into three sentiment categories: Positive, Neutral, and Negative. As a domain-specific model trained on financial text, it effectively captures contextual patterns in financial news.

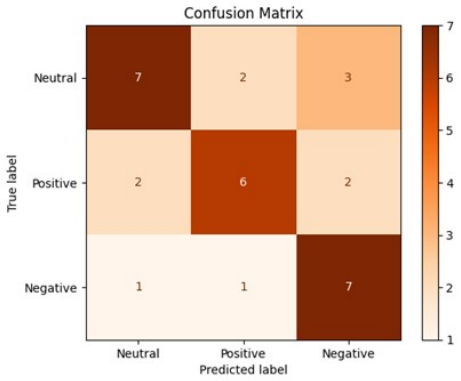

Fig. 5. Confusion Matrix for AMZN (Neutral / Positive / Negative)

The confusion matrix shows the relationship between true and predicted sentiment labels, indicating that FinBERT generates reliable sentiment signals for use in investment decision-support systems.

Table 2. Evaluation of Sentiment Classification

Class	Precision	Recall	F1-Score	Support
Neutral	0.70	0.58	0.64	12
Positive	0.67	0.60	0.63	10
Negative	0.58	0.78	0.67	9

ii) Mapping Sentiment to Trading Signals The classified sentiment outputs were further mapped to corresponding trading signals, including Buy, Hold, and

Sell. While sentiment signals alone cannot fully explain stock price movements, they provide useful directional insights when combined with technical indicators and predictive models.

Table 3. Evaluation of Trading Signals

Class	Precision	Recall	F1-Score	Support
Hold	0.73	0.80	0.76	10
Buy	0.67	0.60	0.63	10
Sell	0.70	0.70	0.70	10

The results indicate that sentiment-derived trading signals provide a meaningful complementary signal for investment decision support. In particular, the model demonstrates strong performance in identifying hold signals, which are important for preventing unnecessary trading during stable market conditions.

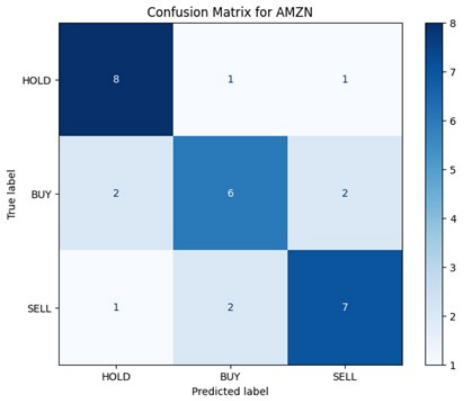

Fig. 6. Confusion Matrix for AMZN Trading Signals (Hold / Buy / Sell)

4.3 Recommendation System

The recommendation system was evaluated through backtesting on 190 Amazon (AMZN) trading instances collected over a one-year period. These samples were excluded from the model training phase to ensure an unbiased assessment of the recommendation module.

The evaluation results indicate that the proposed recommendation framework achieves stable performance across all recommendation categories, with strong performance in identifying hold signals that help avoid unnecessary trading during stable market conditions.

Table 4. Evaluation of Recommendation System

Class	Precision	Recall	F1-Score	Support
Buy	0.76	0.72	0.74	62
Hold	0.82	0.85	0.83	78
Sell	0.74	0.71	0.72	50
Macro Avg	0.77	0.76	0.76	190
Weighted Avg	0.78	0.78	0.78	190

The integration of GA-optimized LSTM price forecasts with FinBERT-based sentiment analysis improves recommendation accuracy by combining quantitative price patterns with qualitative market sentiment. The rule-based framework also ensures transparency and interpretability in the generated investment suggestions.

Overall, the backtesting results demonstrate that the proposed system effectively converts predictive insights into actionable investment recommendations while maintaining consistent performance across different market scenarios.

5 Conclusion

Forecasting stock market behavior remains challenging due to the interaction of quantitative price dynamics and qualitative factors such as financial news and investor sentiment. Relying solely on either source is often insufficient for reliable decision-making. This study proposes a hybrid decision-support framework integrating LSTM-based price prediction, Genetic Algorithm (GA)–based hyperparameter optimization, FinBERT-driven sentiment analysis, and a profile-based recommendation module.

Experimental results show that the GA-optimized LSTM model achieves improved prediction accuracy and stability compared to the baseline model. The inclusion of sentiment signals provides additional contextual insight, while the recommendation module translates predictive outputs into interpretable, user-specific investment guidance. However, the rule-based design limits adaptability to dynamic market conditions and does not explicitly support short-term trading strategies.

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